

Master's Degree in Artificial Intelligence
School of Engineering and Technology

Vector Spaces and Latent Structures in the Mathematical Representation of Documents

Submitted by: Toni Esteve Gene

Supervisor: Dr. Rafael San Miguel Carrasco

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1 Introduction

1.1 Context and motivation

Within the field of artificial intelligence and machine learning, data representation is a fundamental aspect that directly affects the efficiency and effectiveness of predictive models. In particular, when the input data take the form of text, transforming them into a suitable numerical representation poses both a conceptual and a computational challenge (Bengio, Courville, and Vincent 2013, p. 1798).

This issue is especially relevant when one considers that all supervised machine-learning algorithms are ultimately based on computational processes involving numerical data. Consequently, the automatic processing of natural language requires mechanisms capable of transforming textual documents into vectors while preserving the relevant information contained in the text and, at the same time, enabling computationally efficient operations.

The motivation for this study is twofold. First, it arises from an interest in understanding, from a mathematical perspective, the mechanisms through which textual documents are projected into vector spaces so that statistical and machine-learning methods can be applied. Second, this study draws on several subjects completed as part of the Master's Degree in Artificial Intelligence, whose content provided the conceptual tools needed to rigorously address techniques such as *singular value decomposition* (Beltrami 1873) and the *Gibbs sampling method* (S. Geman and D. Geman 1984), both of which are central to some of the representations analysed in this study.

Furthermore, the growing influence of artificial intelligence across many areas of society has highlighted the need to formalise the theoretical foundations of these techniques more thoroughly, moving beyond the superficial treatment with which they are often presented in the scientific literature (Früh and Haux 2022, p. 4).

1.2 Problem statement

The central problem addressed in this study is how textual documents can be transformed into suitable vector representations for subsequent use in machine-learning models. Although numerous techniques exist for representing text numerically, they are not all based on the same mathematical foundations, nor do they exhibit the same computational, predictive, or interpretative behaviour.

In this context, it is necessary to analyse not only the practical operation of these techniques but also their formal foundations. In particular, this study focuses on four relevant approaches to document vector representation: *TF-IDF* (Salton, Wong, and C.-S. Yang 1975), *latent semantic analysis (LSA)* (Deerwester, Susan T Dumais, et al. 1990a), *latent Dirichlet allocation (LDA)* (D. Blei, A. Ng, and M. Jordan 2003), and neural representations based on *Skip-gram with Negative Sampling (SG-NS)* (mikolov2013negative; Mikolov et al. 2013).

Each of these techniques follows a different rationale. TF-IDF is based on the statistical weighting of terms within the vector space model; LSA introduces an algebraic reduction in dimensionality through singular value decomposition and the identification of latent structures; LDA proposes a probabilistic approach in which documents are represented as mixtures of latent topics; and SG-NS learns dense word

representations through a context-based predictive task, which can subsequently be aggregated to construct document vectors.

Therefore, the problem is not limited to determining which technique achieves the best results in a classification task. It also involves understanding why each representation behaves in a particular way from mathematical, computational, and interpretative perspectives. In this regard, it is particularly relevant to compare sparse, high-dimensional representations such as TF-IDF with latent or dense representations such as LSA, LDA, and SG-NS, analysing how each captures textual information, the computational cost involved in constructing it, and the extent to which it facilitates the interpretation of the represented documents and terms.

1.3 Objectives of the study

The general objective of this study is to analyse, formalise, and compare different text vector-representation techniques from both theoretical and empirical perspectives. More specifically, three classical approaches—*TF-IDF*, LSA, and LDA—are studied alongside a neural representation based on *Skip-gram with Negative Sampling* (SG-NS), which is used to construct word embeddings and dense document representations through aggregation strategies.

The following specific objectives are derived from this general objective:

- To mathematically formalise the vector representation of documents using TF-IDF, describing its main components and its interpretation within the vector space model.
- To study the algebraic foundations of latent semantic analysis, paying particular attention to the role of singular value decomposition and dimensionality reduction.
- To describe the probabilistic foundations of latent Dirichlet allocation, analysing its interpretation as a generative document model based on latent topics.
- To analyse the neural foundations of representations based on *Skip-gram with Negative Sampling*, studying how word embeddings are learned from word–context relationships and how they can be aggregated to obtain document vector representations.
- To empirically compare the performance of TF-IDF, LSA, LDA, and SG-NS in binary text-classification problems using supervised-learning models.
- To evaluate the impact of corpus type on each representation by considering different textual settings, such as short and informal texts versus longer and more structured scientific documents.
- To analyse the relationship between predictive performance, complexity, computational cost, dimensionality, and interpretability for each of the representations studied.

1.4 Contributions of the study

The main contributions of this study are divided into two complementary lines. First, it presents a precise formalisation of several techniques used to transform text into vectors: *TF-IDF*, *latent semantic analysis (LSA)*, *latent Dirichlet allocation (LDA)*, and neural representations based on *Skip-gram with Negative Sampling (SG-NS)*. This formalisation includes a description of their underlying mathematical foundations, properties, and computational behaviour.

Although the study builds on established results, it provides an original presentation that examines the mathematical development of the representations in greater depth. For TF-IDF, its statistical basis within the vector space model is analysed; for LSA, the algebraic basis associated with singular value decomposition is developed; for LDA, the probabilistic formulation based on latent topics is studied; and for SG-NS, the neural perspective of learning embeddings from word–context relationships and negative sampling is introduced. In this way, the study combines statistical, algebraic, probabilistic, and neural approaches within a single comparative framework.

The study also analyses how word representations learned with SG-NS can be used to construct dense document representations through aggregation strategies such as the TF-IDF-weighted mean. This addition broadens the comparison beyond classical methods by including a neural technique capable of learning distributional relationships between words from the corpus. It also examines the effect of the embedding training domain by comparing representations learned from an external corpus, such as Wikipedia, with embeddings trained directly on the dataset under analysis.

Second, an empirical comparison designed by the author is conducted across several textual datasets using supervised-learning models for binary classification. This analysis not only quantifies the performance of the different representations according to metrics such as *AUC-ROC*, accuracy, and the F1 score, but also considers the dimensionality, complexity, computational cost, and interpretability of each technique.

In addition, the study offers an original perspective by treating *LDA* as a vector-representation technique that can be useful in supervised settings, rather than solely as an unsupervised generative model. Similarly, SG-NS is incorporated not only as a word-embedding model but also as the basis for constructing document vectors that can be evaluated in classification tasks. Overall, the study provides an integrated comparison of sparse, latent, probabilistic, and neural representations, analysing their advantages and limitations according to corpus characteristics, predictive performance, and associated computational cost.

1.5 Structure of the document

The remainder of the study is organised as follows. First, the general foundations of document vector representation are presented by introducing the concepts of corpus, vocabulary, term–document matrix, and vector space model. The mathematical formalisation of the classical techniques under analysis—TF-IDF, LSA, and LDA—is then developed, with particular attention to their statistical, algebraic, and probabilistic foundations.

The neural representation of language through the *Skip-gram with Negative Sampling* (SG-NS) model is introduced next. This part describes how word embeddings are learned from word–context relationships and how those embeddings can be aggregated to construct dense document vector representations, for example through TF-IDF-weighted means.

The experimental methodology used to compare the different representations in binary text-classification tasks is then described. This part includes the datasets used, the textual preprocessing applied, the implementations developed, the supervised models considered, and the evaluation metrics employed. The results are subsequently presented and analysed by comparing the predictive performance, dimensionality, complexity, computational cost, and interpretability of each representation.

Finally, the implications of the results are discussed from both theoretical and practical perspectives, and the general conclusions of the study are presented together with its main limitations and possible directions for future research.

2 Theoretical framework and state of the art

2.1 TF-IDF

To transform textual documents into a numerical representation that can be processed efficiently by machine-learning algorithms, a vector encoding of those documents is essential. This transformation requires the documents to be structured in advance through a corpus and its corresponding dictionary, as well as the definition of a function that projects each document into a finite-dimensional space. Once these foundations have been established, this section introduces the representation known as *Term Frequency–Inverse Document Frequency* (*TF-IDF*), one of the most widely used techniques in text processing because of its conceptual simplicity and ease of implementation. This description follows the criteria and theoretical foundations presented by (Schütze, Manning, and Raghavan 2008), which provide the basis for the discussion developed below.

2.2 Latent semantic analysis

Latent semantic analysis (*LSA*) is a technique used for information retrieval in text whose objective is to describe latent relationships between words and documents. The method attempts to represent associations among different terms according to their joint occurrence in different contexts. These representations suggest the implicit presence of thematic groups and semantic networks, through which words can be organised according to their conceptual and thematic similarity.

In the following sections, the theoretical foundations required to understand the semantic vector space model used in *LSA* will be developed (Leydesdorff and Welbers 2011; Deerwester, Susan T Dumais, et al. 1990a; Landauer and Susan T Dumais 1997). First, the definition and structure of a semantic vector space will be formally introduced, specifying how the document corpus and vocabulary are represented within that space. The construction of the input matrix, which contains term weights in relation to the documents of a specific corpus, will then be presented. Finally, the decomposition of this matrix through *singular value decomposition* (*SVD*)

and *Perron–Frobenius theory* will be analysed in detail, examining the relevance of this decomposition for reducing dimensionality and obtaining an efficient and meaningful representation of the latent semantic content of the texts.

2.3 Latent Dirichlet allocation

Latent Dirichlet allocation (LDA) is a generative probabilistic model used to represent documents through a latent topic structure. Unlike purely frequency-based representations such as *TF-IDF*, or algebraic dimensionality-reduction techniques such as *LSA*, *LDA* is based on the assumption that each document in a corpus can be understood as a probabilistic combination of topics and that each topic can, in turn, be described as a probability distribution over the words in the vocabulary (D. Blei, A. Ng, and M. Jordan 2003, p. 996).

From this perspective, a document is represented not only by the terms it explicitly contains, but also by the proportions in which different latent topics contribute to its generation. Thus, if a number of topics K is fixed, each document d can be associated with a probability vector

$$\theta_d = (\theta_{d,1}, \theta_{d,2}, \dots, \theta_{d,K}),$$

where each component $\theta_{d,k}$ represents the proportion or weight of topic k within document d . This vector constitutes a compact representation of the document in a K -dimensional topic space, thereby allowing *LDA* to be used as a text vector-representation technique.

The generative nature of *LDA* means that the model describes a probabilistic process through which the observed documents could have been produced. In general terms, the model assumes that a topic distribution is first selected for each document; a topic is then selected for each word in the document according to that distribution; finally, a word is generated from the word distribution corresponding to the selected topic. This formulation allows documents to be interpreted as mixtures of topics and topics as probabilistic groupings of related words (Jelodar et al. 2019, p. 5).

In this way, *LDA* is especially useful when the objective is not only to obtain a numerical representation of the texts but also to discover an underlying thematic structure in the corpus. This property gives it an interpretative advantage over other methods because the resulting latent dimensions can be understood as topics rather than merely as algebraic combinations of terms. For example, in a news corpus, one topic might be associated with sports-related words, another with politics, and another with economics, allowing each document to be represented according to the intensity with which it participates in each of these topics.

Nevertheless, the effectiveness of *LDA* depends heavily on the corpus having a sufficiently clear thematic structure. In very short or noisy texts, or in texts with an unstable vocabulary, the model may have greater difficulty identifying coherent topics. Therefore, although *LDA* provides a compact and interpretable representation, its performance may vary substantially depending on the nature of the corpus and on the selected number of topics K , which acts as a model hyperparameter.

In later sections, this intuition will be formalised mathematically through a description of the *LDA* generative process, the joint probability of the model, the posterior-inference problem, and the use of *Gibbs sampling* as a strategy for estimating latent topic assignments.

2.4 Skip-gram with Negative Sampling

Neural representations based on *embeddings* constitute an alternative approach to classical methods of text vector representation. Unlike techniques such as *TF-IDF*, *LSA*, or *LDA*, which construct representations from frequencies, co-occurrences, or latent structures inferred from term–document matrices, neural models learn dense word vectors through a predictive task. In this context, the *Skip-gram with Negative Sampling* (*SG-NS*) model has become one of the most influential formulations for obtaining distributional word representations ([mikolov2013negative](#); Mikolov et al. 2013).

The fundamental idea of *Skip-gram* is to learn a vector representation of each word from the contexts in which it occurs. Given a corpus viewed as a sequence of terms, the model takes a central word and attempts to predict the words that occur around it within a specified context window. Thus, two words tend to obtain similar vectors when they appear in similar contexts, even if they are not identical and do not necessarily occur together in the same documents. This hypothesis is related to the distributional principle of language, according to which the meaning of a word can be approximated from its contexts of use.

From a vector perspective, the result of training *SG-NS* is an embedding matrix

$$E \in \mathbb{R}^{|V| \times d},$$

where $|V|$ represents the vocabulary size and d the dimension of the latent space. Each row of this matrix corresponds to the dense vector associated with a word in the vocabulary. Therefore, if $w_i \in V$ is a word, its representation is given by a vector

$$v_{w_i} \in \mathbb{R}^d.$$

Unlike a *one-hot* representation, in which each word is encoded as a sparse vector orthogonal to all the others, embeddings learned with *SG-NS* place words in a continuous space where geometric proximity may reflect distributional or semantic similarities.

The use of *negative sampling* is primarily motivated by computational considerations. In the original *Skip-gram* formulation, calculating the probability of a context word requires normalisation over the entire vocabulary through a *softmax* function, which is expensive when $|V|$ is large. Negative sampling transforms this problem into a binary-classification task: the model learns to distinguish word–context pairs observed in the corpus from artificially generated pairs. Thus, rather than updating all vocabulary terms at each training step, only the positive pair and a small number of negative examples are updated, making training substantially more efficient.

However, *SG-NS* directly learns representations of words rather than complete documents. Consequently, when the model is used for document-classification tasks, an aggregation strategy must be defined to combine the embeddings of the words in each document. One simple possibility is to calculate the mean of the word vectors in the document. However, this strategy assigns the same weight to all terms regardless of their importance. A more informative alternative is therefore to use a mean weighted by the *TF-IDF* values, so that the most relevant words in the document exert a greater influence on the final document vector.

Formally, if a document D consists of a sequence of words

$$D = (w_1, w_2, \dots, w_m),$$

and each word w_i has an associated embedding v_{w_i} , the weighted document representation can be expressed as

$$v_D = \frac{\sum_{i=1}^m \text{tfidf}(w_i, D) v_{w_i}}{\sum_{i=1}^m \text{tfidf}(w_i, D)}.$$

In this way, the document is represented by a dense vector in the same latent space as the words while also incorporating information about the statistical relevance of each term within the corpus.

This representation offers several advantages. First, it considerably reduces dimensionality compared with methods based directly on the vocabulary, because the final document dimension is determined by d , rather than by $|V|$. Second, it can capture distributional relationships between words that are not visible in purely frequency-based representations. However, its effectiveness depends heavily on the corpus used to train the embeddings. If the vectors are learned from an external corpus such as Wikipedia, they may capture general semantic relationships but may not reflect the specific uses found in the domain under analysis. Conversely, if they are trained on the dataset itself, they may adapt more closely to the particular language of the corpus, although their quality will depend on the size, contextual richness, and linguistic stability of that dataset.

Consequently, *SG-NS* introduces into the analysis a family of dense neural representations that complement the classical methods studied previously. Whereas *TF-IDF* provides a sparse representation based on the statistical importance of terms, *LSA* constructs a latent space through algebraic reduction, and *LDA* represents documents as probabilistic mixtures of topics, *SG-NS* learns word vectors through a context-based predictive task. Its inclusion in this study therefore makes it possible to compare statistical, algebraic, probabilistic, and neural approaches within a single experimental framework.

3 Mathematical foundations of the techniques analysed

3.1 Corpus, vocabulary, and term–document matrix

3.1.1 Dictionaries and document corpora

Definition 1. *The document corpus is defined as the set of all textual documents under analysis (Schütze, Manning, and Raghavan 2008, p. 19):*

$$D = \{d_1, d_2, \dots, d_n\}$$

where each d_i represents a document in the corpus.

Definition 2. *Given a corpus of textual documents $D = \{d_1, d_2, \dots, d_n\}$, the dictionary is defined as the set of all unique terms present in the corpus (Schütze,*

Manning, and Raghavan 2008, p. 22):

$$\mathcal{V} = \bigcup_{d \in D} T_d$$

where T_d represents the set of terms present in document d .

Given a corpus of textual documents $D = \{d_1, d_2, \dots, d_n\}$, the dictionary $\mathcal{V}$ is understood as a finite set of terms that is usually selected according to a frequency criterion. In most cases, words that do not provide relevant information for the analysis, such as articles and pronouns, are removed; these words are known as *stopwords*. Formally, a threshold $k \in \mathbb{N}$ is established to represent the maximum number of terms to be included in the dictionary. This threshold normally becomes a hyperparameter to be tuned when a supervised-learning model is applied:

$$\mathcal{V}_k = \{t_1, t_2, \dots, t_k\}, \quad \text{where } t_i \text{ is one of the } k \text{ most frequent terms in } D.$$

3.1.2 Term–document matrix

Using the previous definition of a semantic vector space model, consider a document corpus $\mathcal{D}$ containing $|\mathcal{D}| = n$ documents and an associated vocabulary $\mathcal{V}$ containing $|\mathcal{V}| = m$ terms. Based on this representation, the documents are organised into an input matrix $A \in \mathbb{R}^{m \times n}$, where each column corresponds to the vector $\mathbf{v}_j = \phi(d_j)$ representing document $d_j \in \mathcal{D}$ within the semantic vector space previously defined through the mapping function ϕ . The entries a_{ij} of this matrix are determined using weighting functions such as *TF-IDF*, thereby reflecting the relevance of term t_i in document d_j . This matrix representation therefore captures both the global importance of terms across the entire document collection and their local specificity within each individual document (Leydesdorff and Welbers 2011, p. 3).

$$\begin{array}{c} \\ t_1 \\ t_2 \\ \vdots \\ t_i \\ \vdots \\ t_m \end{array} \begin{bmatrix} d_1 & d_2 & \dots & d_j & \dots & d_n \\ a_{11} & a_{12} & \dots & a_{1j} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2j} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{ij} & \dots & a_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix}$$

Figure 1: Schematic representation of the input matrix A , where the rows correspond to terms t_i and the columns to documents d_j . *Source: Author's own work.*

In general, the input matrix A usually has a sparse structure because each document typically uses only a small subset of the terms available in the vocabulary. This follows from the fact that documents generally address specific topics, which naturally limits the number of terms used in each one.

3.2 Mathematical formalisation of TF-IDF

TF-IDF (Term Frequency–Inverse Document Frequency) is a function that assigns a weight to each term in a document according to its frequency in that document and its rarity within a document corpus.

Definition 3. Let $D = \{d_1, d_2, \dots, d_n\}$ be a document corpus and let $\mathcal{V}$ be the vocabulary of terms present in D . The term frequency (TF) of a term $t \in \mathcal{V}$ in a document $d \in D$ is defined as (Schütze, Manning, and Raghavan 2008, p. 117):

$$tf(t, d) = \frac{f(t, d)}{\max\{f(t', d) \mid t' \in d\}}$$

where:

- $f(t, d)$ is the number of times term t occurs in document d .
- $\max\{f(t', d) \mid t' \in d\}$ is the maximum frequency of any term in document d .
- **Purpose:** To normalise the term frequency with respect to the most frequent words in the document and thereby avoid bias towards long texts.

Definition 4. The inverse document frequency (IDF) is defined as (Schütze, Manning, and Raghavan 2008, p. 118):

$$idf(t, D) = \log \left(\frac{|D|}{|\{d \in D \mid t \in d\}|} \right)$$

where:

- $|D|$ is the total number of documents in the corpus.
- $|\{d \in D \mid t \in d\}|$ is the number of documents in which term t occurs.
- **Purpose:** To penalise terms that appear in many documents because they provide less discriminative information.

Definition 5. The TF-IDF weight of a term t in a document d is defined as (Schütze, Manning, and Raghavan 2008, p. 119):

$$tfidf(t, d) = tf(t, d) \times idf(t, D)$$

- This value combines the local importance of the term within a document (TF) with its global importance in the corpus (IDF).
- Terms that occur frequently in a document but are rare in the corpus receive a high TF-IDF value, making them more relevant for classification and text analysis.

In a TF-IDF problem, the documents in the corpus $D = \{d_1, d_2, \dots, d_n\}$ are represented in a vector space of dimension $|\mathcal{V}|$, where $\mathcal{V}$ is the term dictionary.

The vector representation of a document d_i is defined as:

$$\mathbf{v}_{d_i} = (tfidf(t_1, d_i), tfidf(t_2, d_i), \dots, tfidf(t_{|\mathcal{V}|}, d_i)) \in \mathbb{R}^{|\mathcal{V}|}$$

where:

- Each component of the vector $\mathbf{v}_{d_i}$ corresponds to the *TF-IDF* weight of term $t_j \in \mathcal{V}$ in document d_i .
- The length of the vector is $|\mathcal{V}|$, which is the total number of terms in the dictionary.

This numerical representation of the documents facilitates the application of machine-learning models.

3.3 Algebraic foundations of LSA

3.3.1 Semantic vector space model

Definition 6. Let $\mathcal{D} = \{d_1, d_2, \dots, d_n\}$ be a document corpus and let $\mathcal{V}$ be a vocabulary of size m . A semantic vector space is defined as a pair $(\mathbb{R}^m, \phi)$, where $\mathbb{R}^m$ is an m -dimensional Euclidean vector space in which each document is represented as a vector, and $\phi : \mathcal{D} \rightarrow \mathbb{R}^m$ is a mapping function that maps each document to a vector in $\mathbb{R}^m$.

Each element $d_i \in \mathcal{D}$ is represented by a vector $\mathbf{v}_i \in \mathbb{R}^m$, whose structure depends on the representation method used (Berry and Browne 2005, p. 4). One of the most popular representations is based on *TF-IDF*, as explained previously.

This structure makes it possible to perform algebraic operations to measure similarities and extract latent information from a large document corpus D . Through this representation, it can also be assumed that the meaning of the documents may be derived directly from the vocabulary $\mathcal{V}$ from which they are composed.

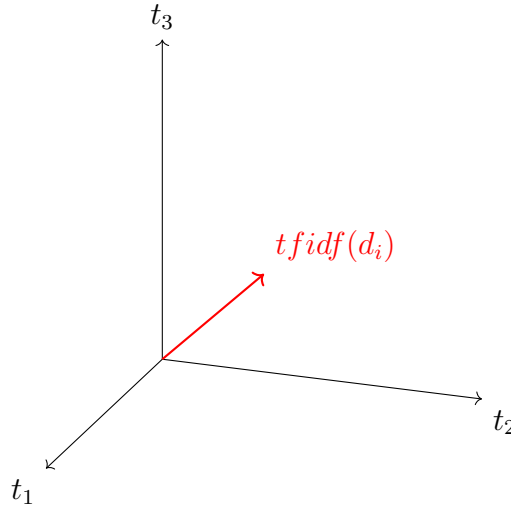


Figure 2: Vector representation of a document in a vocabulary consisting of three terms. *Source: Author's own work.*

3.3.2 Decomposition of the input matrix into orthogonal components

From an algebraic perspective, the matrix A defined previously contains information about how terms are associated with documents and how documents share terms. This twofold relationship means that the relationships between terms and documents

are not independent, but rather correlated. In other words, the occurrence of a particular term in a document is often conditioned by the presence of other terms, and documents can be grouped according to common semantic patterns.

These interdependencies are algebraically reflected in the fact that the matrix A exhibits linear or nearly linear dependencies between rows and/or columns. For example, given the vector representation of the word *beach* across the documents, it is highly likely that it will be very similar to the vector representation of the word *sea*. They are also reflected in a correlation structure that is visible in the matrices $A^T A$ and AA^T , which respectively indicate similarities between documents and between terms.

This situation suggests the existence of an underlying latent structure, that is, a reduced set of latent semantic factors that explain term co-occurrence and document similarity. However, this structure is not captured in the original representation of A , because the documents and terms are expressed in canonical bases that do not explicitly reflect these deeper conceptual relationships (Landauer and Susan T Dumais 1997, p. 214).

It is precisely in this context that singular value decomposition (SVD) becomes a fundamental tool. Let us recall the result established by (Dongarra et al. 2018, p. 810):

Theorem 1. (*Singular Value Decomposition*) *Let $A \in \mathbb{R}^{m \times n}$ be a real matrix of dimension $m \times n$. Then, there exist two orthogonal matrices $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$, and a diagonal matrix $\Sigma \in \mathbb{R}^{m \times n}$, such that:*

$$A = U\Sigma V^T$$

where:

- U is an orthogonal matrix, that is, $U^T U = I_m$.
- V is an orthogonal matrix, that is, $V^T V = I_n$.
- Σ is a diagonal matrix of the form:

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{m \times n}$$

where $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ are the singular values of A , and $r = \text{rank}(A)$.

- The first r columns of U are the left singular vectors, that is, the eigenvectors of AA^T .
- The first r columns of V are the right singular vectors, that is, the eigenvectors of $A^T A$.

- The singular values σ_i are the non-negative square roots of the non-zero eigenvalues of $A^T A$, or equivalently of AA^T .

The *SVD* $A = U\Sigma V^T$ provides an orthogonal decomposition that separates and restructures the latent relationships between terms and documents through three transformations. On the one hand, V identifies orthogonal combinations of documents representing independent semantic dimensions in the input space; Σ quantifies the relevance of each of these dimensions through the singular values σ_i ; and U expresses how these dimensions are manifested in the term space, defining a new orthogonal basis for the output space.

This decomposition shows that $Av_i = \sigma_i u_i$, that is, each orthogonal combination of documents v_i is transformed by A into an orthogonal combination of terms u_i , scaled by σ_i . In this way, the *SVD* diagonalises the relationship between documents and terms in a latent space, providing a decoupled and structured representation that reveals the essential components of the semantic content of the corpus (Deerwester, Susan T Dumais, et al. 1990a, p. 398).

3.4 Singular value decomposition and dimensionality reduction

3.4.1 Truncation of the Singular Value Decomposition

In high-dimensional and sparse settings, as is commonly the case in document analysis, the *SVD* makes it possible to identify a set of latent dimensions that capture underlying semantic patterns between terms and documents. However, not all these dimensions contribute significantly to the representation of the content. As will be shown, dimensions associated with the smallest singular values often capture irrelevant variation or statistical noise.

For this reason, a common practice in such cases is to apply a truncation of the *SVD*, which consists of removing the latent dimensions with the lowest weights and retaining only the first k most informative components. This reduction not only simplifies the model and improves its interpretability, but also preserves most of the relevant semantic variability of the corpus. It therefore makes it possible to discover hidden semantic connections by grouping terms and documents with similar meanings into shared conceptual dimensions, facilitating the identification of synonyms and improving information retrieval without requiring exact word matches (Martin and Berry 2007, p. 42).

From the *SVD*, a matrix $A \in \mathbb{R}^{m \times n}$, where $r = \text{rank}(A)$, can be written as

$$A = U_r \Sigma_r V_r^T,$$

where $U_r = [u_1, u_2, \dots, u_r] \in \mathbb{R}^{m \times r}$ is a matrix containing the first r columns of U , which are mutually orthogonal; $V_r = [v_1, v_2, \dots, v_r] \in \mathbb{R}^{n \times r}$ is a matrix containing the first r columns of V , which are also mutually orthogonal; and $\Sigma_r \in \mathbb{R}^{r \times r}$ is a diagonal matrix containing the non-zero singular values of A ,

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0.$$

Thus, if the dimensionality of the matrix A is reduced by selecting only the first k most significant components, where $k < r$, the concepts associated with the smallest singular values, which are less important in the representation of the model, are removed. This produces the matrix

$$A_k = U_k \Sigma_k V_k^T,$$

where $U_k = [u_1, u_2, \dots, u_k] \in \mathbb{R}^{m \times k}$ is a matrix containing the first k columns of U , which are mutually orthogonal; $V_k = [v_1, v_2, \dots, v_k] \in \mathbb{R}^{n \times k}$ is a matrix containing the first k columns of V , which are mutually orthogonal; and $\Sigma_k \in \mathbb{R}^{k \times k}$ is a diagonal matrix containing the k largest singular values of A ,

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k > 0$$

(Horasan et al. 2019, p. 2).

The theoretical foundation of this approximation is provided by the *Eckart–Young–Mirsky Theorem*, which guarantees that the matrix A_k is the best rank- k approximation to the original matrix A , minimising either the 2-norm or the Frobenius norm¹:

Theorem 2. (Eckart and Young 1936, p. 211)

Let

$$D = U \Sigma V^\top \in \mathbb{R}^{m \times n}, \quad m \geq n,$$

be the singular value decomposition (SVD) of D , where $\Sigma := \text{diag}(\sigma_1, \dots, \sigma_m)$ is the rectangular diagonal matrix containing the singular values

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_m.$$

For $r \in \{1, \dots, m-1\}$, partition U , Σ , and V as follows:

$$U = [U_1 \quad U_2], \quad \Sigma = \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix}, \quad V = [V_1 \quad V_2],$$

where U_1 has size $m \times r$, Σ_1 has size $r \times r$, and V_1 has size $n \times r$.

Then, the rank- r matrix obtained from the truncated singular value decomposition is defined as

$$\hat{D}^* = U_1 \Sigma_1 V_1^\top,$$

such that

$$\|D - \hat{D}^*\|_F = \min_{\text{rank}(\hat{D}) \leq r} \|D - \hat{D}\|_F = \sqrt{\sigma_{r+1}^2 + \dots + \sigma_m^2}.$$

The minimiser $\hat{D}^*$ is unique if and only if $\sigma_{r+1} \neq \sigma_r$. See the proof in (Eckart and Young 1936, p. 216).

¹The Frobenius norm of A , denoted by $\|A\|_F$, is defined as:

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}.$$

The theorem not only states that A_k is the solution to the problem

$$\min_{\text{rank}(B) \leq k} \|A - B\|_F,$$

but also that

$$\|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \sigma_{k+2}^2 + \cdots + \sigma_r^2}.$$

In other words, the singular values σ_i determine the *amount of information* contained in each direction. By truncating the *SVD* to the k most important dimensions, the contribution of the directions associated with

$$\sigma_{k+1}, \sigma_{k+2}, \dots, \sigma_r$$

is removed. This contribution is precisely the information lost in the best rank- k approximation.

3.4.2 Semantic connections and latent structure through Perron–Frobenius theory

From a conceptual perspective, *LSA* has been shown to identify latent relationships between documents and terms by grouping them into latent concepts that capture the maximum variance of the information contained in the original matrix A , as established by the *Eckart–Young Theorem* [2].

Nevertheless, it is natural to ask why this method is capable of revealing semantic relationships between documents that do not explicitly share any term, but nevertheless address thematically similar content. Likewise, it is useful to understand how synonymous terms that do not co-occur in any document in the corpus can ultimately be represented together within the same latent structure.

This phenomenon can be rigorously justified through the application of *Perron–Frobenius theory*, particularly within the analysis of non-negative matrices and the distinction between reducible and irreducible matrices. This theory provides the structural conditions under which the semantic relationships between documents, or between terms, can either be decomposed into disjoint blocks in the reducible case or exhibit a strongly connected structure in the irreducible case. This distinction has direct implications for the ability of *LSA* to capture global semantic relationships through singular value decomposition (Valle-Lisboa and Mizraji 2007, p. 15).

Let $A \in \mathbb{R}^{m \times n}$ be the term–document matrix, where each column represents a document $d_j \in \mathbb{R}^m$, and each row represents a term in the vocabulary V . Define the matrix

$$D = A^\top A \in \mathbb{R}^{n \times n},$$

which measures similarity between documents in the term space. Each entry D_{ij} of the matrix D is the inner product between the vectors d_i and d_j , that is,

$$D_{ij} = \langle d_i, d_j \rangle = \sum_{k=1}^m A_{k,i} A_{k,j}.$$

This quantity measures the degree of co-occurrence of the vocabulary terms $\mathcal{V}$ between documents d_i and d_j . Therefore, D acts as a document-similarity matrix based on the information distributed across the terms, which will be referred to as the *co-occurrence matrix*.

If the singular value decomposition of A is considered,

$$A = U\Sigma V^\top,$$

the diagonalisation of D is obtained as

$$D = A^\top A = (U\Sigma V^\top)^\top (U\Sigma V^\top) = V\Sigma^\top U^\top U\Sigma V^\top = V\Sigma^2 V^\top. \quad (1)$$

We now introduce several key concepts from *Perron–Frobenius theory* (Meyer 2000, p. 674), which establishes special properties concerning the eigenvalues and eigenvectors of reducible and irreducible non-negative matrices.

Definition 7. (*Reducible and irreducible matrices*)

Let $D \in \mathbb{R}^{n \times n}$ be a square matrix. The matrix D is said to be reducible if there exists a permutation matrix $P \in \mathbb{R}^{n \times n}$ such that

$$P^\top D P = \begin{pmatrix} X & Y \\ 0 & Z \end{pmatrix},$$

where $X \in \mathbb{R}^{k \times k}$ and $Z \in \mathbb{R}^{(n-k) \times (n-k)}$ are square matrices for some $1 \leq k < n$, and 0 denotes the zero matrix of size $(n-k) \times k$.

Otherwise, D is said to be irreducible.

Definition 8. (*Decomposable matrix*)

A matrix $D \in \mathbb{R}^{n \times n}$ is said to be decomposable, or block diagonal, if there exists a permutation matrix $P \in \mathbb{R}^{n \times n}$ such that D can be expressed as a block-diagonal matrix:

$$P^\top D P = \begin{pmatrix} D_1 & 0 & \cdots & 0 \\ 0 & D_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & D_m \end{pmatrix},$$

where each D_i is a square matrix and all off-diagonal blocks are zero.

Definition 9. Let $A = (a_{ij}) \in \mathbb{R}^{n \times n}$ be a matrix. The directed graph associated with A , denoted by $\mathcal{G}(A)$, is defined as the graph with node set

$$\{N_1, N_2, \dots, N_n\},$$

in which there is a directed edge from N_i to N_j if and only if

$$a_{ij} \neq 0.$$

Definition 10. The graph $\mathcal{G}(A)$ is said to be strongly connected if, for every pair of nodes (N_i, N_k) , there exists a directed path from N_i to N_k ; that is, there exists a sequence of directed edges connecting N_i to N_k .

In fact, the following characterisation can be established.

Lemma 1. *A matrix $A \in \mathbb{R}^{n \times n}$ is irreducible if and only if the graph $\mathcal{G}(A)$ is strongly connected. See the proof in (Meyer 2000, p. 742).*

Theorem 3. *(Perron–Frobenius Theorem for irreducible non-negative matrices)*

Let $D \in \mathbb{R}^{n \times n}$ be an irreducible and non-negative matrix, that is, $D \geq 0$. Then, the following properties hold:

- *The spectral radius $r = \rho(D)$, defined as the maximum absolute value of the eigenvalues of D , is real, positive, and has algebraic multiplicity equal to 1.*
- *There exists an eigenvector $\mathbf{v} \in \mathbb{R}^n$ such that $\mathbf{v} > 0$ and*

$$D\mathbf{v} = r\mathbf{v}.$$

- *There exists a left eigenvector $\mathbf{p} \in \mathbb{R}^n$, with $\mathbf{p} > 0$ and $\|\mathbf{p}\|_1 = 1$, such that*

$$\mathbf{p}^\top D = r\mathbf{p}^\top.$$

This vector $\mathbf{p}$ is called the Perron vector, and the pair $(r, \mathbf{p})$ is called the Perron pair.

No other non-negative eigenvector of D is admissible, except for positive multiples of $\mathbf{p}$, regardless of the associated eigenvalue. See the proof in (Khim 2007, p. 2).

Let $D = A^\top A$ be the document co-occurrence matrix obtained from the term-document matrix A . By construction, D is symmetric and non-negative, because each entry D_{ij} corresponds to the inner product between the vectors representing documents d_i and d_j .

Because D is symmetric, the *Spectral Theorem*² guarantees that it can be diagonalised using an orthonormal basis and that all its eigenvalues are real, as shown in Equation (1). This property facilitates the application of the *Perron–Frobenius Theorem* [3], which studies the spectral behaviour of non-negative matrices.

Now interpret D as the adjacency matrix of a directed graph $G(D)$, where each node represents a document and there is a directed edge $d_i \rightarrow d_j$ whenever $D_{ij} \neq 0$, that is, whenever the documents share at least one term. In this context, the irreducibility of D is equivalent to the graph being strongly connected [10]. This means that, for every pair of documents (d_i, d_j) , there is a path, potentially passing through intermediate documents, connecting d_i to d_j , and vice versa (Valle-Lisboa and Mizraji 2007, p. 17).

Applying the *Perron–Frobenius Theorem* [3] to the matrix D shows that, if D is irreducible, its dominant eigenvalue $\rho(D)$ is strictly positive and there exists an

²*Spectral Theorem*, real symmetric matrix version: Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Then, there exists an orthonormal basis of eigenvectors of A . Equivalently, there exists an orthogonal matrix $Q \in \mathbb{R}^{n \times n}$, where $Q^\top Q = I$, such that

$$A = Q\Lambda Q^\top,$$

where Λ is a diagonal matrix containing the eigenvalues of A , all of which are real.

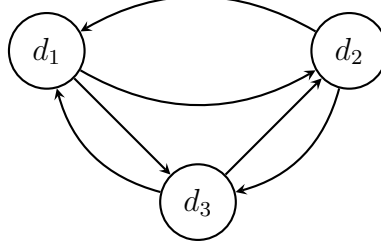


Figure 3: Example of a strongly connected directed graph associated with an adjacency matrix D , where each node represents a document and the edges indicate the existence of shared terms. The fact that the graph is strongly connected guarantees the irreducibility of D , ensuring that a directed path exists between every pair of documents. *Source: Author’s own work.*

associated strictly positive eigenvector $\mathbf{v}_1 > 0$. In this context, $\mathbf{v}_1$ reflects the presence of a global latent structure that semantically connects all the documents, even when they do not explicitly share the same terms.

The contrapositive of this result implies that, if the dominant eigenvector contains zero coordinates, this indicates that D is reducible. Consequently, there are subsets of documents that are semantically disconnected from the others within the latent space: they either share no terms with the remainder of the corpus or there is no co-occurrence chain connecting them.

In general, the existence of a strictly positive eigenvector for a non-negative matrix does not imply that the matrix is irreducible. However, this statement does hold in the particular case in which the matrix is symmetric. As shown in the following lemma, the fact that the dominant eigenvector of D is strictly positive guarantees that the associated co-occurrence graph is strongly connected. In the present context, this means that all documents are connected through some semantic path within the latent structure represented by D .

Lemma 2. *Let $D = A^\top A$ be a symmetric and non-negative matrix of dimension $n \times n$. Let σ_1 be the dominant eigenvalue of D , that is, the eigenvalue with maximum absolute value. Suppose that σ_1 has algebraic multiplicity 1 and that there exists an eigenvector $\mathbf{v}_1 \in \mathbb{R}^n$ associated with σ_1 such that $\mathbf{v}_1 > 0$, meaning that all its components are strictly positive.*

Then, D is irreducible; equivalently, the associated directed graph $\mathcal{G}(D)$ is strongly connected. See the proof in (Valle-Lisboa and Mizraji 2007, p. 18).

Although the co-occurrence matrix $D = A^\top A$ is irreducible in some cases, meaning that all documents are semantically connected through a co-occurrence path, in practice D may be reducible. This situation occurs when the document collection contains completely disconnected thematic subsets, that is, groups of documents that do not share any terms with the remainder of the corpus.

From an algebraic perspective, the reducibility of D is manifested through the appearance of exact zeros in the first singular vector of A , indicating the presence of orthogonal components within the latent space. In this setting, the following property makes it possible to interpret the structure as a decomposition into semantically independent blocks, each of which can be analysed separately.

Lemma 3. *Let $D_{n \times n}$ be a symmetric matrix. If D is reducible, then it is decomposable, that is, it can be written as a block-diagonal matrix after a suitable permutation.*

For the matrix $D = A^T A$, this indicates that the corpus contains mutually unrelated subsets of documents: groups of documents that share no terms with other groups. Each resulting diagonal block corresponds to an independent thematic cluster, representing a coherent grouping of documents dealing with the same topic but semantically disconnected from the other clusters.

This observation implies that the directed graph $\mathcal{G}(D)$ associated with the co-occurrence matrix D has several strongly connected components, each corresponding to an autonomous thematic block within the corpus. The documents within each component are semantically interconnected, whereas no co-occurrence path connects them to documents belonging to other components (Valle-Lisboa and Mizraji 2007, p. 20).

3.4.3 Vector representation of documents and similarities

One of the main advantages of latent semantic analysis is that it not only represents documents as vectors in a reduced-dimensional space, but also facilitates the calculation of similarity measures between terms or between documents without requiring direct operations on the original high-dimensional matrix.

By projecting the data into the latent space generated by the truncated singular value decomposition, comparisons can be performed efficiently and accurately in a low-dimensional orthonormal subspace. This preserves the relevant semantic information while significantly reducing computational cost.

Let $d_i \in \mathbb{R}^m$ be the vector representation of a document obtained from its *TF-IDF* weights, and let $e_i \in \mathbb{R}^n$ be the canonical vector associated with the i -th column, so that

$$d_i = A e_i.$$

Considering the truncated decomposition

$$A_k = U_k \Sigma_k V_k^T,$$

the following approximation is obtained:

$$d_i = A e_i \approx A_k e_i = U_k \Sigma_k v_i,$$

where $v_i \in \mathbb{R}^k$ is the i -th row of V_k^T , expressing the projection of document d_i onto the k principal latent components.

Therefore, if only the representation of the document in the k -dimensional latent space is retained, the following vector is obtained (Martin and Berry 2007, p. 49):

$$\hat{d}_i = v_i = \Sigma_k^{-1} U_k^T d_i \in \mathbb{R}^k.$$

This vector constitutes the reduced and semantically meaningful version of d_i in terms of the k latent concepts identified by the truncated *SVD*. The same expression can be applied generally to project any document, whether from the training set, as in the preceding derivation, or from the test set, into the reduced semantic space. This guarantees a coherent and comparable representation within the constructed latent model.

3.5 Probabilistic foundations of LDA

Having introduced *latent Dirichlet allocation* from a conceptual perspective, this section formally describes its mathematical operation. The objective is to explain how *LDA* models the generation of documents from latent variables, how the joint distribution of the model is expressed, and how the topic representation of each document can be estimated using approximate inference procedures.

3.5.1 The LDA generative process

The generative process in *LDA* describes how a document in a corpus is generated under the assumption that each document consists of a mixture of latent topics and that each word in the document originates from one of those topics (D. Blei, A. Ng, and M. Jordan 2003, p. 996). Let a dictionary $\mathcal{V}$ be defined from the document corpus D , and let K be a fixed number of latent topics in the corpus.

For each latent topic $k = 1, \dots, K$, a distribution over all words in the vocabulary is assigned:

$$\phi^{(k)} = \left(\phi_1^{(k)}, \phi_2^{(k)}, \dots, \phi_{|\mathcal{V}|}^{(k)} \right) \sim \text{Dirichlet}(\beta).$$

The vector $\phi^{(k)}$ represents the probability distribution of the words within topic k . Therefore, each component $\phi_j^{(k)}$ indicates the probability that the j -th word in the vocabulary will occur when the selected latent topic is k .

Similarly, for each document d , the model assumes a topic distribution given by

$$\theta_d = (\theta_{d,1}, \theta_{d,2}, \dots, \theta_{d,K}) \sim \text{Dirichlet}(\alpha),$$

where $\theta_{d,k}$ represents the proportion of topic k in document d . Thus, the document is represented as a probability vector over the K latent topics.

The steps of the *LDA* generative process for each document $\mathbf{w}$ in a corpus D are as follows:

1. Determine the length of the document. The number of words N in the document is selected according to a Poisson distribution:

$$N \sim \text{Poisson}(\zeta).$$

2. Select the topic distribution for the document. A topic distribution θ is selected for the document from a *Dirichlet* distribution with parameter α :

$$\theta \sim \text{Dirichlet}(\alpha).$$

3. Generate the words in the document. For each word w_n :

- A topic z_n is selected from a multinomial distribution based on θ :

$$z_n \sim \text{Multinomial}(\theta).$$

- A word w_n is then selected from a multinomial distribution determined by the selected topic z_n :

$$w_n \sim \text{Multinomial}(\phi^{(z_n)}).$$

Thus, *LDA* defines a hierarchical model in which documents are generated from topic distributions and words are generated from distributions associated with those topics.

3.5.2 Joint probability of the LDA model

The joint probability rule establishes how the joint probability of a collection of random variables can be calculated from their conditional probabilities.

Lemma 4. (*Joint probability rule*)

For a set of n random variables $X_1, X_2, \dots, X_n$, the joint probability can be decomposed as

$$p(X_1, X_2, \dots, X_n) = p(X_1)p(X_2 | X_1)p(X_3 | X_1, X_2) \dots p(X_n | X_1, \dots, X_{n-1}).$$

See the proof in (Ferrante and Bardina 2023, p. 6).

Given a document d containing N words, the objective is to model the joint probability of all the variables involved in the generative process:

$$p(\theta, \mathbf{z}, \mathbf{w}, \phi | \alpha, \beta).$$

Applying the joint probability rule gives

$$p(\theta, \mathbf{z}, \mathbf{w}, \phi | \alpha, \beta) = p(\phi | \beta, \alpha)p(\theta | \phi, \alpha, \beta)p(\mathbf{z} | \theta, \phi, \alpha, \beta)p(\mathbf{w} | \mathbf{z}, \theta, \phi, \alpha, \beta).$$

Taking the conditional dependencies of the model into account, the word distribution for a topic depends only on β , and therefore

$$p(\phi | \alpha, \beta) = p(\phi | \beta).$$

Likewise, the topic distribution within a document depends only on the hyperparameter α , so that

$$p(\theta | \phi, \alpha, \beta) = p(\theta | \alpha).$$

Moreover, given θ , the topic assignments $\mathbf{z}$ are selected according to a multinomial distribution:

$$p(\mathbf{z} | \theta, \alpha, \phi, \beta) = p(\mathbf{z} | \theta).$$

Finally, the selection of each word depends on its assigned topic and on the word distribution associated with that topic:

$$p(\mathbf{w} \mid \mathbf{z}, \theta, \phi, \alpha, \beta) = p(\mathbf{w} \mid \mathbf{z}, \phi).$$

Therefore, the joint distribution of the *LDA* model is given by (Carpenter 2010, p. 2):

$$p(\phi, \theta, \mathbf{z}, \mathbf{w} \mid \alpha, \beta) = p(\phi \mid \beta)p(\theta \mid \alpha)p(\mathbf{z} \mid \theta)p(\mathbf{w} \mid \mathbf{z}, \phi). \quad (2)$$

This expression summarises the fundamental probabilistic dependencies of the model. First, the word distributions for each topic are generated; then, the topic distribution for each document is generated; afterwards, latent topic assignments are made for the words; and finally, the observed words are generated.

3.5.3 Posterior inference in LDA

As observed in the *LDA* generative process, the model postulates the existence of an underlying structure of latent topics that organises the content of the documents. Within this framework, each document can be represented by a probability vector θ , which encodes the relative proportion with which each topic contributes to the generation of the observed content.

The inference problem in *LDA* consists of calculating the posterior distribution of the hidden variables θ , $\mathbf{z}$, and ϕ , given an observed document $\mathbf{w}$. The objective is therefore to compute

$$p(\theta, \mathbf{z}, \phi \mid \mathbf{w}, \alpha, \beta).$$

Applying Bayes' rule gives (Darling 2011, p. 3):

$$p(\theta, \mathbf{z}, \phi \mid \mathbf{w}, \alpha, \beta) = \frac{p(\theta, \mathbf{z}, \mathbf{w}, \phi \mid \alpha, \beta)}{p(\mathbf{w} \mid \alpha, \beta)}.$$

The denominator is the marginal probability of the observed documents and is obtained by integrating over the latent variables:

$$p(\mathbf{w} \mid \alpha, \beta) = \int \int p(\theta \mid \alpha) \prod_{i=1}^N \sum_{\mathbf{z}} p(\phi \mid \beta) p(z_n \mid \theta) p(w_n \mid z_n, \phi) d\theta d\phi.$$

In practice, this expression is computationally intractable, particularly when the number of documents, words, and topics is large. Consequently, approximate inference methods must be used, among which *Gibbs sampling* is especially prominent.

3.5.4 Gibbs sampling

The posterior distribution formulated above can be approximated using a method known as *Gibbs sampling*. This method belongs to the family of Markov chain Monte Carlo algorithms (*MCMC*) and makes it possible to obtain samples from a complex joint distribution using its full conditional distributions.

As described by (Gelfand 2000, p. 1301), the central idea is to use knowledge of the conditional distributions of the components of a random vector

$$\mathbf{x} = (x_1, \dots, x_m)$$

to construct a Markov chain whose stationary distribution is the target distribution

$$p(x_1, \dots, x_m).$$

Suppose that samples are required from a random vector $\mathbf{x} = (x_1, \dots, x_m)$ with joint distribution $p(x_1, \dots, x_m)$, which is difficult to calculate directly. If the following conditional distributions are known,

$$p(x_1 \mid x_2, \dots, x_m),$$

$$p(x_2 \mid x_1, x_3, \dots, x_m),$$

and so forth, Gibbs sampling proceeds iteratively as follows:

1. Each x_i is randomly initialised.
2. For $n = 1, \dots, N$, the components are sequentially updated:

- Sample $x_1^{(n+1)}$ using

$$p(x_1 \mid x_2^{(n)}, x_3^{(n)}, \dots, x_m^{(n)}).$$

- Sample $x_2^{(n+1)}$ using

$$p(x_2 \mid x_1^{(n+1)}, x_3^{(n)}, \dots, x_m^{(n)}).$$

- Continue the process until x_m has been updated.

After a sufficiently large number of iterations, the generated samples approximate samples drawn from the target joint distribution.

3.5.5 Collapsed Gibbs sampling in LDA

In the specific case of *LDA*, inference focuses on three principal latent variables: the document-topic distribution θ_d , the topic-word distribution $\phi^{(k)}$, and the topic assignments z_i for each word.

Because the posterior distributions of θ and ϕ can be integrated analytically under the conjugate Dirichlet–multinomial structure, the inference process can be simplified using *collapsed Gibbs sampling*. Under this approach, neither θ nor ϕ is sampled explicitly; only the topic assignments

$$\mathbf{z} = (z_1, \dots, z_N)$$

are sampled.

This strategy reduces the dimensionality of the sampling space and improves the computational efficiency of inference (Xiao and Stibor 2010, p. 63).

The objective is to iteratively calculate the conditional probability that a particular word is assigned to a given topic, conditional on the assignments of all the other words. Formally, the probability of interest is

$$p(z_i = k \mid z_{-i}, \mathbf{w}, \alpha, \beta),$$

where z_{-i} represents all topic assignments except the assignment associated with word i .

The collapsed Gibbs sampling equation for *LDA* is given by (Darling 2011, p. 6):

$$p(z_i = k \mid z_{-i}, \mathbf{w}, \alpha, \beta) \propto \frac{n_{d,k}^{(-i)} + \alpha_k}{\sum_j (n_{d,j}^{(-i)} + \alpha_j)} \cdot \frac{n_{k,w}^{(-i)} + \beta_w}{\sum_{w'} (n_{k,w'}^{(-i)} + \beta_{w'})}.$$

In this expression:

- $n_{d,k}^{(-i)}$ denotes the number of words in document d assigned to topic k , excluding the current word i .
- $n_{k,w}^{(-i)}$ denotes the number of times word w has been assigned to topic k , excluding the current word.
- α_k controls the topic distribution within each document.
- β_w controls the word distribution within each topic.

The first factor measures the probability that document d uses topic k , whereas the second factor measures the probability that topic k generates the observed word w . Thus, the assignment of each word to a topic simultaneously depends on the topic composition of the document and the lexical distribution of each topic.

3.5.6 Point estimation of the parameters θ_d and ϕ_k

Once the latent topic assignments for the words have been obtained through *collapsed Gibbs sampling*, point estimates can be calculated for the model parameters that describe the vector representation of the documents and the word distribution of each topic (Heinrich 2005, p. 23).

For each document d , the topic distribution θ_d can be estimated as

$$\hat{\theta}_{d,k} = \frac{n_{d,k} + \alpha_k}{\sum_{j=1}^K (n_{d,j} + \alpha_j)}.$$

This expression gives the estimated probability that document d concerns topic k . Therefore, the vector

$$\hat{\theta}_d = (\hat{\theta}_{d,1}, \hat{\theta}_{d,2}, \dots, \hat{\theta}_{d,K})$$

constitutes the vector representation of the document in the latent topic space generated by *LDA*.

Similarly, for each topic k , the word distribution ϕ_k can be estimated as

$$\hat{\phi}_{k,w} = \frac{n_{k,w} + \beta_w}{\sum_{w'=1}^V (n_{k,w'} + \beta_{w'})}.$$

This expression gives the estimated probability that word w is generated conditional on topic k . Thus, each topic is characterised by a probability distribution over the vocabulary.

3.5.7 Training an LDA model

After completing an iteration of Gibbs sampling, the hyperparameters α and β , which respectively control the topic distribution within documents and the word distribution within topics, can be updated (Minka 2000, p. 1).

The parameter α can be adjusted by maximising the marginal distribution of the topic assignments:

$$p(\mathbf{z} \mid \boldsymbol{\alpha}) = \prod_{d=1}^D \frac{B(n_{d,\cdot} + \boldsymbol{\alpha})}{B(\boldsymbol{\alpha})}.$$

Taking the log-likelihood of the preceding expression gives

$$\log p(\mathbf{z} \mid \boldsymbol{\alpha}) = \sum_{d=1}^D \log \frac{B(n_{d,\cdot} + \boldsymbol{\alpha})}{B(\boldsymbol{\alpha})}.$$

Since

$$\log B(\boldsymbol{\alpha}) = \sum_{k=1}^K \log \Gamma(\alpha_k) - \log \Gamma\left(\sum_{k=1}^K \alpha_k\right),$$

the following expression is obtained:

$$\begin{aligned} \log p(\mathbf{z} \mid \boldsymbol{\alpha}) &= \sum_{d=1}^D \left[\sum_k \log \Gamma(n_{d,k} + \alpha_k) - \log \Gamma\left(\sum_k (n_{d,k} + \alpha_k)\right) \right] \\ &\quad - D \left[\sum_k \log \Gamma(\alpha_k) - \log \Gamma\left(\sum_k \alpha_k\right) \right]. \end{aligned}$$

Differentiating with respect to α_k and setting the result equal to zero gives

$$\frac{\partial}{\partial \alpha_k} \log p(\mathbf{z} \mid \boldsymbol{\alpha}) = \sum_d \left[\psi(n_{d,k} + \alpha_k) - \psi\left(\sum_j (n_{d,j} + \alpha_j)\right) \right] - D \left[\psi(\alpha_k) - \psi\left(\sum_j \alpha_j\right) \right] = 0, \quad (3)$$

where $\psi(x)$ is the digamma function defined as

$$\psi(x) = \frac{\partial}{\partial x} \log \Gamma(x).$$

The preceding expression does not admit a closed-form solution, and numerical methods must therefore be used. In this regard, (Minka 2000, p. 2) proposes reformulating the expression as a fixed-point equation and shows that the Newton–Raphson method exhibits local convergence under certain conditions. The same reasoning can be applied analogously to the parameter β .

3.6 Skip-gram with Negative Sampling

3.6.1 Transition towards neural language representations

Classical text vector-representation techniques such as *TF-IDF*, *LSA*, and *LDA* transform documents into numerical vectors using different mathematical principles.

However, bag-of-words-based models have several limitations because they tend to discard term order and may represent certain semantic relationships between words inadequately. In contrast, neural language models learn distributed and continuous representations that generalise more effectively to unseen sequences and assign nearby vectors to semantically related words or documents (Le and Mikolov 2014, p. 2).

3.6.2 The *Skip-gram* model

The *Skip-gram* model is one of the classical neural formulations for learning dense word vector representations. Its objective is to exploit local co-occurrence relationships between terms. Given a central word, the model attempts to predict the words that occur in its context within a specified window. Consequently, embedding learning is formulated as a predictive task based on the relationship between central words and context words.

Let a corpus be represented as a sequence of words

$$w_1, w_2, \dots, w_T,$$

and let $c \in \mathbb{N}$ be the context-window size. For each central word w_t , the context words are those in positions w_{t+j} such that

$$-c \leq j \leq c, \quad j \neq 0.$$

The objective of *Skip-gram* is to maximise (Mikolov et al. 2013)

$$\sum_{t=1}^T \sum_{\substack{-c \leq j \leq c \\ j \neq 0}} \log P(w_{t+j} \mid w_t).$$

Therefore, the model learns by assigning high probabilities to the words that actually occur in the context of each central word.

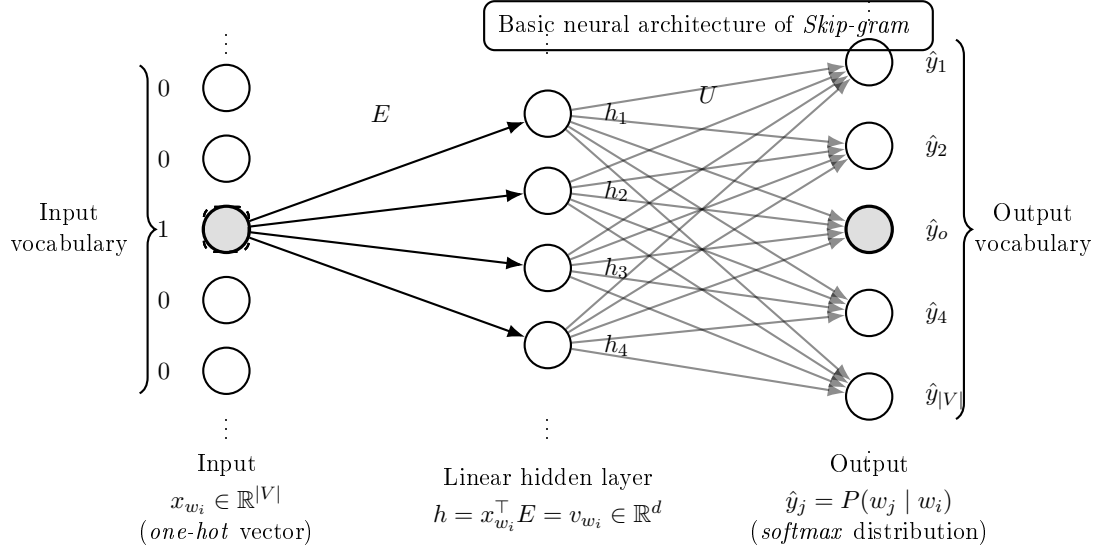


Figure 4: Diagram of the neural architecture of *Skip-gram*, based on (Rong 2014, p. 8).

3.6.3 Neural architecture of *Skip-gram*

From the perspective of classical neural architectures, the *Skip-gram* model can be interpreted as a particular case of a multilayer perceptron (*MLP*) with a single linear hidden layer and zero bias terms (Rong 2014, p. 7).

Indeed, if the convention is adopted whereby the *one-hot* vector of the central word is represented as a column vector

$$x_{w_i} \in \mathbb{R}^{|V|},$$

the first transformation of the network can be written as

$$h = E^\top x_{w_i} + b^{(1)}.$$

In the basic formulation of *Skip-gram*, no bias term is introduced in the hidden layer, so that

$$b^{(1)} = 0.$$

Moreover, the activation function of this layer is the identity function. Therefore,

$$h = E^\top x_{w_i} = v_{w_i}.$$

This shows that the first layer of the model does not perform a non-linear transformation. Instead, it performs a linear projection that selects the embedding associated with the central word.

The output layer then calculates a score vector over the entire vocabulary:

$$z = Uh + b^{(2)}.$$

Again, the output bias term is omitted in the classical formulation of the model:

$$b^{(2)} = 0.$$

Consequently,

$$z = Uh = UE^\top x_{w_i}.$$

Each component z_j measures the compatibility between the central word w_i and a possible context word w_j :

$$z_j = u_{w_j}^\top v_{w_i}.$$

Finally, the *softmax* function transforms these scores into a probability distribution (Mikolov et al. 2013, p. 7):

$$P(w_j \mid w_i) = \frac{\exp(z_j)}{\sum_{k=1}^{|V|} \exp(z_k)}. \quad (4)$$

Therefore, *Skip-gram* can be regarded as a degenerate or simplified MLP whose only trainable parameters are the matrices

$$\theta = \{E, U\},$$

with zero bias terms

$$b^{(1)} = 0, \quad b^{(2)} = 0,$$

and an identity activation function in the hidden layer. The distinctive feature of the model is that the hidden layer is interpreted not merely as an intermediate transformation, but as the dense vector representation of the central word.

3.6.4 From *Skip-gram* to *Skip-gram with Negative Sampling*

The basic formulation of *Skip-gram* makes it possible to interpret embedding learning as a predictive task in which, given a central word w_i , the model must assign a high probability to the words appearing in its context. In the full *softmax* version, this probability is defined by Equation (4).

This expression is conceptually clear because it defines a probability distribution over all the words in the vocabulary. However, it presents a major computational difficulty, since the denominator requires calculating the inner product

$$u_w^\top v_{w_i}$$

for every word $w \in V$. Therefore, each model update requires traversing the entire vocabulary, whose size $|V|$ may be very large in real-world corpora (Rong 2014).

From a computational perspective, the cost of evaluating the full *softmax* is of the order

$$\mathcal{O}(|V|d),$$

where d is the embedding dimension. When $|V|$ contains tens or hundreds of thousands of terms, this calculation becomes too expensive to perform for every observed word–context pair (Mikolov et al. 2013, p. 3). For this reason, a more efficient approximation known as *negative sampling* is introduced (Mikolov et al. 2013, p. 3).

The fundamental idea of *negative sampling* is to reformulate the original *Skip-gram* problem. Instead of posing a multiclass classification problem over the entire vocabulary, in which the model would have to identify the correct context word among all the possible words in V , the task is transformed into a binary classification problem. In this way, the model no longer attempts to select the correct word directly from the whole vocabulary, but rather determines whether a word–context pair actually comes from the corpus or whether it has been artificially generated as a negative sample (Mikolov et al. 2013, p. 3).

Let (w_i, w_o) be a positive pair, where w_i is the central word and w_o is a word that actually appears within its context window. The model also generates S negative words

$$w_1^-, w_2^-, \dots, w_S^-,$$

sampled from a noise distribution $P_n(w)$ ³. These negative words are not interpreted as observed contexts of w_i , but as artificial examples against which the model must learn to discriminate (Mikolov et al. 2013, pp. 3–4).

Under this formulation, the objective function for a positive pair (w_i, w_o) can be written as

$$\ell(w_i, w_o) = \log \sigma(u_{w_o}^\top v_{w_i}) + \sum_{s=1}^S \log \sigma(-u_{w_s^-}^\top v_{w_i}),$$

where

$$\sigma(x) = \frac{1}{1 + \exp(-x)}$$

is the sigmoid function (Mikolov et al. 2013, pp. 3–4).

The first term,

$$\log \sigma(u_{w_o}^\top v_{w_i}),$$

corresponds to the positive pair. Maximising this term encourages

$$u_{w_o}^\top v_{w_i}$$

to be large. In other words, the model adjusts the vectors so that the central word w_i and its observed context w_o have high compatibility.

By contrast, each negative term

³In *negative sampling*, $P_n(w)$ is the distribution used to sample negative words, that is, words presented to the model as artificial or unobserved contexts for a given central word. In the original formulation by Mikolov et al., the unigram distribution raised to the power 3/4 is used.

$$\log \sigma(-u_{w_s}^\top v_{w_i})$$

encourages

$$u_{w_s}^\top v_{w_i}$$

to be small or negative. In other words, the model learns to reduce the compatibility between the central word and the artificially generated contexts (Goldberg and Levy 2014, p. 5).

In this way, *Skip-gram with Negative Sampling* transforms the original training process into a discriminative task in which pairs observed in the corpus must receive a high score, whereas negative pairs must receive a low score. The computational cost per update no longer depends on the entire vocabulary, but only on the number of negative samples:

$$\mathcal{O}((S+1)d),$$

where normally

$$S \ll |V|.$$

Therefore, *negative sampling* makes it possible to train embedding models much more efficiently when working with large vocabularies (Rong 2014, pp. 13–14).

It should be noted that this modification mainly affects the objective function and the way in which the model output is calculated, but it does not alter the initial structure of the neural architecture. Indeed, the input remains the *one-hot* vector of the central word $x_{w_i} \in \mathbb{R}^{|V|}$, and the first transformation still consists of a linear projection through the embedding matrix E . Therefore, the hidden layer is obtained in the same way as in the basic *Skip-gram* model:

$$h = E^\top x_{w_i} = v_{w_i}.$$

This hidden layer includes neither a bias term nor a nonlinear activation function, and can therefore be interpreted as a linear layer whose function is to select the input embedding of the central word. The difference introduced by *negative sampling* appears after this representation. Instead of projecting h over the entire vocabulary and normalising the result through a *softmax*, the model calculates only the scores associated with the positive context and a small number of sampled negative contexts. Thus, the architecture retains the same intermediate representation v_{w_i} , but changes the way in which the output is trained, replacing multiclass prediction over the entire vocabulary with a binary task that discriminates between real and artificial word–context pairs.

3.6.5 Vector representation of documents from embeddings

However, the *Skip-gram* model directly learns vector representations of words rather than complete documents. Once the model has been trained, an embedding matrix is obtained:

$$E \in \mathbb{R}^{|V| \times d},$$

where each row E_i contains the dense representation of a word $w_i \in V$:

$$v_{w_i} = E_i \in \mathbb{R}^d.$$

Therefore, the direct output of the model is a mapping

$$\varphi : V \longrightarrow \mathbb{R}^d, \quad \varphi(w_i) = v_{w_i},$$

which assigns each word in the vocabulary a vector in the latent space. However, when the objective is to represent complete documents, an additional aggregation function must be defined to combine the embeddings of the words that make up each document.

Although *Skip-gram* directly learns word representations, these representations can subsequently be aggregated to construct document vectors. A common strategy consists of combining the word vectors in the document by means of an arithmetic mean or a weighted mean using relevance measures such as IDF or TF-IDF (Singh and Mukerjee 2015, p. 15). These document vectors can then be used as input features for a supervised classifier (Singh and Mukerjee 2015, p. 17).

Let

$$D = (w_1, w_2, \dots, w_m)$$

be a document consisting of m words. Given the embeddings

$$v_{w_1}, v_{w_2}, \dots, v_{w_m} \in \mathbb{R}^d,$$

a first approximation consists of representing the document using the arithmetic mean of its word vectors:

$$v_D = \frac{1}{m} \sum_{i=1}^m v_{w_i}.$$

In this way, the document is represented by a vector

$$v_D \in \mathbb{R}^d,$$

located in the same latent space as the words. This representation can be interpreted as a global summary of the lexical content of the document, constructed from the centroid of the embeddings of its terms.

However, the arithmetic mean assigns the same weight to every word in the document, regardless of its informational relevance. To introduce differentiated weighting, a weight function can be defined as

$$\alpha_i = \alpha(w_i, D),$$

where $\alpha_i \geq 0$ measures the importance of the word w_i within the document D . In this case, the document representation can be written in general form as a weighted mean:

$$v_D = \frac{\sum_{i=1}^m \alpha_i v_{w_i}}{\sum_{i=1}^m \alpha_i}.$$

A common choice is to use the TF-IDF values of each word in the document as weights:

$$\alpha_i = \text{tfidf}(w_i, D).$$

Thus, the following representation is obtained:

$$v_D = \frac{\sum_{i=1}^m \text{tfidf}(w_i, D) v_{w_i}}{\sum_{i=1}^m \text{tfidf}(w_i, D)}.$$

This formulation allows the most informative words to exert a greater influence on the final representation, whereas highly frequent or weakly discriminative terms contribute to a lesser extent.

From the perspective of a supervised document-classification task, the *Skip-gram* model can be used as a preliminary unsupervised-learning stage aimed at obtaining dense word vector representations. Once the embedding matrix E has been learned, each document D_i in the dataset can be transformed into a vector $v_{D_i} \in \mathbb{R}^d$ using an aggregation function

$$g : (\mathbb{R}^d)^m \longrightarrow \mathbb{R}^d,$$

such that

$$v_{D_i} = g(v_{w_1}, v_{w_2}, \dots, v_{w_m}).$$

These document vectors can subsequently be used as input to a supervised classifier. If y_i denotes the label associated with document D_i , a classification model can be trained:

$$F_\beta : \mathbb{R}^d \longrightarrow \mathcal{Y},$$

such that

$$\hat{y}_i = F_\beta(v_{D_i}),$$

where β represents the parameters of the classifier and $\mathcal{Y}$ is the set of possible classes.

Consequently, the use of *Skip-gram* for document classification can be understood as a two-stage procedure. First, a distributed representation of the words is learned from an unsupervised context-prediction task. Second, the documents are represented by aggregating these embeddings and are then used as input to a supervised model. Therefore, although *Skip-gram* does not directly generate document embeddings, its word representations can serve as the basis for constructing dense document vectors.

3.6.6 Further analysis of embedding geometry in *Skip-gram with Negative Sampling*

The geometric interpretation of embeddings should not be understood solely as a free arrangement of words in a Euclidean space in which proximity expresses semantic similarity. In the specific case of *Skip-gram with Negative Sampling* (SGNS), the geometry of the latent space is conditioned by the model's objective function itself. Therefore, the final positions of the vectors depend not only on the semantic regularities of the corpus, but also on the balance between positive and negative examples introduced during training (Mimno and Thompson 2017, p. 2873).

SGNS learns two families of vectors. On the one hand, it learns word vectors

$$v_w \in \mathbb{R}^d,$$

which are the embeddings normally retained at the end of training. On the other hand, it learns context vectors

$$u_c \in \mathbb{R}^d,$$

associated with words when they act as contexts. Thus, for a central word w and a context word c , the model considers the inner product

$$v_w^\top u_c,$$

which measures the compatibility between the two representations (Mimno and Thompson 2017, p. 2873).

In the negative-sampling formulation, the model not only attempts to increase compatibility between observed pairs (w, c) , but also to reduce compatibility between the central word w and randomly sampled negative contexts c_s . A common way of expressing the objective for a positive pair (w, c) and S negative samples is

$$\ell(w, c) = \log \sigma(v_w^\top u_c) + \sum_{s=1}^S \log \sigma(-v_w^\top u_{c_s}),$$

where

$$\sigma(x) = \frac{1}{1 + \exp(-x)}$$

is the sigmoid function (Mimno and Thompson 2017, p. 2874).

The first term,

$$\log \sigma(v_w^\top u_c),$$

encourages the inner product between a word and an observed context to be large. Geometrically, this tends to bring the word vector v_w closer to the context vector u_c , or at least to make both vectors point in compatible directions. By contrast, the negative terms,

$$\log \sigma(-v_w^\top u_{c_s}),$$

encourage the inner product between v_w and the negative contexts u_{c_s} to be small or negative. Geometrically, this introduces a repulsive force between word vectors and negative context vectors (Mimno and Thompson 2017, pp. 2874, 2876).

This tension is fundamental to understanding the resulting geometry of the embedding space (Mimno and Thompson 2017, p. 2876).

From this perspective, if two words w_i and w_j appear in similar contexts, both must exhibit high compatibility with similar sets of context vectors. In other words, for many contexts c_r , the model tends to impose relationships of the form

$$v_{w_i}^\top u_{c_r} \approx v_{w_j}^\top u_{c_r}.$$

An efficient way to satisfy these constraints is for

$$v_{w_i} \quad \text{and} \quad v_{w_j}$$

to have similar orientations in the latent space. Therefore, words with similar context distributions tend to be represented by nearby vectors or vectors with high cosine similarity (Mimno and Thompson 2017, pp. 2873–2874):

$$\cos(v_{w_i}, v_{w_j}) = \frac{v_{w_i}^\top v_{w_j}}{\|v_{w_i}\| \|v_{w_j}\|}.$$

However, the geometric analysis of SGNS shows that this interpretation requires some qualification. Not all vectors are uniformly distributed throughout the space $\mathbb{R}^d$. In particular, word embeddings tend to occupy a relatively concentrated region of the space, which can be interpreted as a narrow cone around a principal direction. If the mean embedding vector is defined as

$$\bar{v} = \frac{1}{|V|} \sum_{w \in V} v_w,$$

then the fact that many inner products

$$v_w^\top \bar{v}$$

are positive indicates that many vectors point approximately towards the same region of the space, rather than being uniformly distributed around the origin (Mimno and Thompson 2017, pp. 2873–2875).

This phenomenon implies that embedding geometry cannot be interpreted solely in terms of local semantic proximity. A global structure induced by the training objective also exists. In SGNS, word vectors and context vectors tend to occupy opposite regions of the latent space. In simplified terms, the optimisation favours a configuration in which

$$v_w^\top u_c < 0$$

for many unobserved or negative word–context combinations. This global separation allows the model to distinguish between real pairs and pairs generated through negative sampling (Mimno and Thompson 2017, pp. 2875–2876).

The number of negative samples S also influences this geometry. When S increases, the relative weight of the negative terms within the objective function also increases. Therefore, the repulsive force associated with the negative contexts becomes more important. Consequently, the number of negative samples affects not only training efficiency, but also the final geometric shape of the vector space (Mimno and Thompson 2017, pp. 2876–2877).

Thus, the geometric representation of embeddings in SGNS should be understood at two levels. At the local level, words with similar contexts tend to have nearby vectors because they must predict similar context distributions. At the global level, the structure of the space is conditioned by the negative-sampling objective, which induces a particular organisation of word and context vectors. Therefore, geometric proximity between embeddings reflects distributional regularities of language, but also the constraints imposed by the loss function used during learning (Mimno and Thompson 2017, p. 2877).

4 Experimental methodology

Having developed the theoretical foundations of the main text vector-representation techniques, this analysis was complemented by a practical contribution aimed at empirically and computationally studying the actual behaviour of the methods formalised above.

4.1 Technical considerations

The experiments were conducted in the *Google Colab* environment using the virtual machines provided by the platform (Bisong 2019; Google 2026). When available, an *NVIDIA Tesla T4 GPU* was used, featuring 16 GB of *GDDR6* memory, 2,560 CUDA cores, and Tensor Cores designed to accelerate matrix operations and deep-learning models (NVIDIA Corporation 2019; NVIDIA Corporation 2018).

However, it was taken into account that some stages of the experiment, such as textual preprocessing, vocabulary construction, or the explicit calculation of frequencies, depended mainly on the CPU and memory management. Therefore, the use of the GPU did not guarantee uniform acceleration throughout the entire process (TensorFlow 2026; Hennessy and Patterson 2019).

4.2 Description of the experiment

The methodology was applied to different textual datasets selected to represent heterogeneous scenarios in terms of size, document length, number of classes, textual noise, and thematic structure. This diversity made it possible to study whether the behaviour of each representation depended on the type of corpus and whether certain techniques were more appropriate for short texts, long documents, thematic corpora, or large-scale collections.

A relevant part of the analysis consisted of constructing the vector representations through original implementations developed from the mathematical formulations described in the theoretical framework. To this end, the main stages of each method were explicitly programmed, allowing the internal operations of each representation to be analysed with greater control.

In addition, the representations obtained through these original implementations were compared with those generated by high-level libraries such as *scikit-learn*. This comparison made it possible to examine the differences in efficiency and performance between an implementation designed to facilitate understanding of the procedure and an implementation optimised for practical use. For this purpose, the times associated with constructing each representation and subsequently training the supervised classifier were recorded.

To evaluate the usefulness of the generated representations, the resulting vectors were used as input to the same supervised-classification model, specifically a *Random Forest* classifier. The use of a single classifier for all representations enabled a homogeneous comparison, preventing performance differences from being attributable to the predictive model rather than to the quality of the vector representation.

4.3 Performance evaluation

Performance was analysed using metrics such as *accuracy*, *precision*, *recall*, the F1 score, and AUC-ROC. This made it possible to assess whether a more complex or computationally expensive representation provided a genuine improvement in the predictive capacity of the classifier, or whether simpler methods offered competitive results at a lower computational cost.

To formally define these metrics, the values obtained from the confusion matrix were considered (Tharwat 2021, p. 3):

- *TP*, or *True Positives*, indicates the number of positive-class examples correctly classified by the model.
- *TN*, or *True Negatives*, indicates the number of negative-class examples correctly classified by the model.
- *FP*, or *False Positives*, indicates the number of negative-class examples incorrectly classified as positive by the model.
- *FN*, or *False Negatives*, indicates the number of positive-class examples incorrectly classified as negative by the model.

Accuracy measures the total proportion of correctly classified observations relative to the total number of examples (Tharwat 2021, p. 4):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

This metric makes it possible to evaluate the overall performance of the classifier, although its interpretation may be limited when the class distribution is imbalanced.

Precision measures the proportion of truly positive examples among all those classified as positive by the model (Tharwat 2021, p. 6):

$$\text{Precision} = \frac{TP}{TP + FP}.$$

A high precision value indicates that the classifier produces few false positives.

Recall, also known as sensitivity or the true positive rate, measures the proportion of positive examples correctly identified relative to the total number of examples that actually belong to the positive class (Tharwat 2021, p. 5):

$$\text{Recall} = \frac{TP}{TP + FN}.$$

A high *recall* value indicates that the model was able to detect most of the positive examples, producing few false negatives.

The F1 score was calculated as the harmonic mean of precision and *recall* (Tharwat 2021, p. 7):

$$\text{F1-Score} = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

This metric provides a balanced measure of classifier performance and is especially useful when the classes are imbalanced or when it is important to consider false positives and false negatives simultaneously.

Finally, *AUC-ROC* evaluates the ability of the classifier to discriminate between classes based on the predicted probabilities. The ROC curve is constructed by plotting the true positive rate against the false positive rate for different decision-threshold values (Tharwat 2021, p. 11).

The true positive rate is defined as

$$\text{TPR} = \frac{TP}{TP + FN},$$

whereas the false positive rate is defined as

$$\text{FPR} = \frac{FP}{FP + TN}.$$

The area under the ROC curve was mathematically expressed as

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR}).$$

The *AUC-ROC* value ranged between 0 and 1. A value close to 1 indicated a high discriminative capacity, whereas a value close to 0.5 represented behaviour similar to random classification. In multiclass problems, these metrics were calculated for each class using a *one-versus-rest* strategy and subsequently aggregated using macro or weighted averages.

4.4 Datasets used

To evaluate text vector representations in diverse scenarios, two textual datasets with clearly heterogeneous characteristics were used: *Sentiment140* (Go, Bhayani, and Huang 2009), designed for sentiment analysis in short Twitter messages, and a dataset containing metadata from scientific articles available on *arXiv* (Cornell University 2026).

These corpora were selected to analyse the behaviour of the studied techniques across different types of language, document structures, and classification tasks, an aspect that is particularly relevant when evaluating textual representation and classification methods (Sebastiani 2002; Aggarwal and Zhai 2012).

4.4.1 Sentiment140

The first dataset considered was *Sentiment140*, which consists of short messages obtained from *Twitter* and was originally designed for sentiment-analysis tasks. The training corpus contains a total of 1,600,000 English-language tweets collected between April and June 2009, evenly distributed between 800,000 messages with positive polarity and 800,000 messages with negative polarity.

Each observation contains six fields: the polarity of the message, the tweet identifier, the publication date, the query through which it was retrieved, the username, and the textual content of the message.

The labels in the training corpus were not assigned manually, but through a strategy based on the presence of positive or negative emoticons. Once this information had been used to determine polarity, the emoticons employed as labels were removed from the messages. This procedure made it possible to construct a large corpus, although it introduced some degree of noise because the presence of an emoticon does not always correctly reflect the overall sentiment expressed in the text.

This corpus is characterised by short, informal, and potentially noisy texts. The original Twitter messages were limited to 140 characters. In addition, they may contain abbreviations, user mentions, URLs, tags or *hashtags*, spelling mistakes, colloquial expressions, repeated characters, and poorly structured syntactic constructions.

For this reason, *Sentiment140* is especially useful for evaluating the robustness of vector representations when documents contain little contextual information and exhibit highly variable vocabulary.

Although the original dataset contains 1,600,000 training records, a balanced sample of 20,000 valid documents was selected for the experiments conducted in this study. Of these, 16,000 were used to train the models and 4,000 were used for evaluation. This reduction kept the computational cost manageable and made it possible to apply the same data partitions to the different representation techniques, thereby ensuring a homogeneous comparison.

4.4.2 arXiv dataset

The second dataset was obtained from a snapshot of the metadata available in *arXiv*, a repository specialising in the dissemination of scientific articles. Unlike *Sentiment140*, the total number of records in this dataset is not constant, because

arXiv adds new papers and updates its metadata on a daily basis. The version downloaded in March 2026 contained approximately three million articles.

Each record represents a scientific article and contains several metadata fields, including the arXiv identifier, authors, title, abstract, subject categories, comments, publication reference, DOI, licence, different document versions, and update date. To construct the textual information used by the models, the title and abstract of each article were primarily considered and were subsequently subjected to the preprocessing procedure defined in this study.

This corpus contains technical, formal, and more highly structured texts characteristic of academic and scientific language. The documents are considerably longer than the messages in *Sentiment140* and exhibit greater conceptual density, specialised terminology, and more carefully organised language.

They may also contain mathematical expressions, acronyms, names of methods, technical references, and infrequent vocabulary, making them a particularly demanding scenario for representations based on limited vocabularies.

Articles in *arXiv* may be associated simultaneously with several subject categories. Therefore, the original structure of the dataset is potentially multilabel and reflects the interdisciplinary nature of many scientific papers. In this study, the first category listed in the metadata was taken as the primary category, and the original problem was transformed into a binary classification task.

Documents whose primary category began with the prefix `cs.` were assigned to the computer-science class, whereas the remaining documents were grouped into the `no_cs` class.

A balanced sample of 100,000 articles was constructed from the preprocessed corpus, consisting of approximately 50,000 computer-science documents and 50,000 documents belonging to other subject areas. This balance prevented classifier performance from being conditioned by the original frequency of the categories.

However, the binary transformation simplified the thematic structure of the dataset because it grouped highly heterogeneous disciplines under a single label and discarded part of the information provided by the secondary categories.

The combined use of *Sentiment140* and *arXiv* made it possible to evaluate the representation techniques in two clearly differentiated scenarios. *Sentiment140* represented a domain characterised by short, informal, noisy texts with limited context, whereas *arXiv* provided longer, formal, specialised, and semantically dense documents.

This heterogeneity made it possible to analyse the extent to which the behaviour of the representations depended on document length, degree of structure, lexical richness, and thematic nature.

4.4.3 Text preprocessing

Before constructing the vector representations, a common preprocessing procedure was applied to all the documents in the datasets used. The objective of this stage was to standardise the textual content, reduce noise, and obtain a linguistic representation better suited to the subsequent construction of the vectors. In this way, it was ensured that the differences observed between the representations were not

due to different text-processing procedures, but rather to the representation method itself.

The preprocessing procedure was systematically applied to the textual column selected in each dataset. For this purpose, the *spaCy* library was used, a natural language processing tool for Python designed for tasks such as tokenisation, linguistic analysis, and text normalisation (Explosion AI 2026). This common preprocessing procedure served as the basis for all the vector representations analysed, whereas the subsequent method-specific transformations depended on the characteristics of each technique.

4.4.4 Exploratory analysis of the datasets

To provide a general characterisation of the corpora used, a brief exploratory analysis was conducted, focusing on the properties most relevant to the construction of vector representations.

Table 1: Exploratory summary of document length before and after preprocessing.

Dataset	Metric	Original text	Preprocessed text	Classes
Sentiment140	Mean number of words	13.25	6.07	2
Sentiment140	Median number of words	12.00	6.00	2
arXiv Dataset	Mean number of words	154.70	91.37	176
arXiv Dataset	Median number of words	151.00	89.00	176

Table 1 and Figures 5 show clear differences between the two datasets used. In the case of *Sentiment140*, the documents are very short, with an average of 13 words in the original text and 6 words after preprocessing. This reduction is consistent with the nature of the corpus, which consists of short tweets in which the removal of URLs, mentions, punctuation marks, *stopwords*, and other non-informative elements has a proportionally large impact. In relative terms, preprocessing reduces the average document length by approximately 54%.

By contrast, the *arXiv Dataset* contains considerably longer documents, with an average of 154.70 words before preprocessing and 91.37 words after preprocessing. Although a substantial reduction is also observed, it is proportionally smaller than in *Sentiment140*, at approximately 41%. This behaviour can be explained by the more formal and structured nature of scientific texts, in which a larger proportion of the content consists of technical and semantically relevant terms that are retained after text cleaning.

The histograms reinforce this difference between the two corpora. Whereas *Sentiment140* concentrates most documents at very short lengths, the *arXiv Dataset* exhibits a distribution shifted towards longer documents. This difference is relevant to the subsequent comparison of vector representations because short texts provide less lexical context and may make it more difficult to identify latent structures, whereas scientific texts provide greater semantic and thematic density.

Finally, it should be noted that the *arXiv Dataset* originally presents a more complex classification structure, organised into 176 main categories in addition to their corresponding subcategories. However, to standardise the experimental design and facilitate comparison between representations, this structure was transformed into a binary classification problem in this study. Specifically, a distinction was made

between documents belonging to the field of computer science, identified through *cs.** categories, and documents belonging to all other fields, thereby defining the task as a classification between *cs* and *non-cs*.

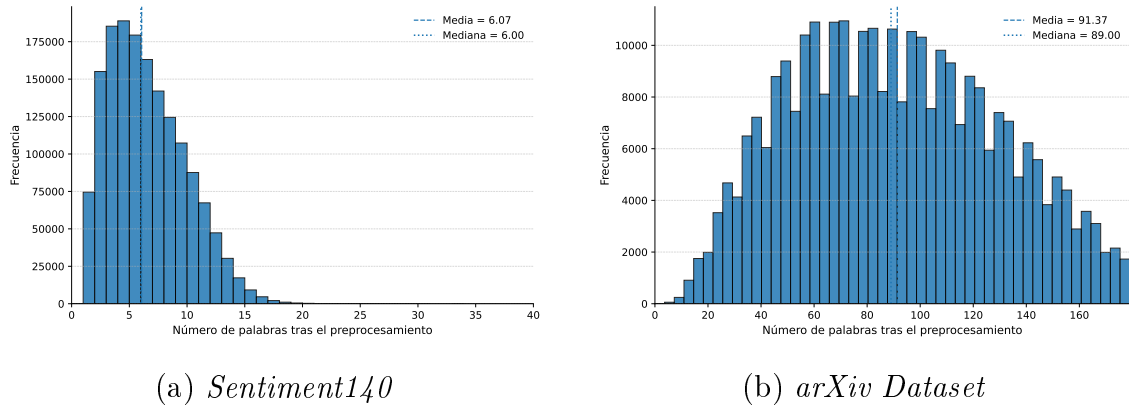


Figure 5: Distribution of preprocessed document lengths in the datasets used. In both cases, length is measured as the number of words remaining after applying text preprocessing.

4.5 Results and discussion

This subsection presents and analyses the results obtained from the experiments conducted on the datasets considered. The analysis is organised separately for each dataset in order to interpret the behaviour of the different vector representations in heterogeneous textual settings. First, the results corresponding to *Sentiment140*, characterised by short, informal texts typical of social media language, are discussed. The results obtained for the *arXiv Dataset*, which consists of more formal, technical, and structured scientific texts, are then analysed.

For each dataset, the discussion considers predictive-performance metrics, execution times, representation dimensionality, computational complexity, and degree of interpretability. Therefore, the discussion is not limited to identifying which method obtains the best results, but also evaluates the strengths and limitations of each representation according to the characteristics of the corpus under analysis.

4.5.1 Results for Sentiment140

This subsection presents the results obtained for the *Sentiment140* dataset. This corpus makes it possible to evaluate the behaviour of the different vector representations in a sentiment-classification setting based on short, informal, and potentially noisy texts typical of the language used on Twitter. The results achieved by each method are analysed below in terms of the predictive performance of the classifier, the computational cost associated with constructing the representation, and the interpretability of the generated vectors.

Predictive performance of the representations The results obtained for the *Sentiment140* dataset when evaluating the different manually implemented vector representations are presented below. To ensure a homogeneous comparison, all the

representations were evaluated using the same data partition and the same supervised classifier, with its parameters held constant. In this way, the observed differences can be attributed primarily to the way in which each method transforms documents into vectors.

Table 2: Predictive and computational results obtained for *Sentiment140* using manual implementations of the vector representations.

Representation	Dim.	Accuracy	F1	AUC-ROC	Repr. time (s)	Clf. time (s)	Sparsity
Manual TF-IDF	2435	0.7250	0.7250	0.7949	0.92	149.02	0.9980
Manual LSA	100	0.7163	0.7162	0.7797	51.46	22.06	0.0144
Manual LDA	10	0.5285	0.5280	0.5335	115.54	1.95	0.0000
Manual SG-NS + TF-IDF	100	0.6148	0.6146	0.6681	332.28	44.83	0.0000
Manual Sentiment 140 SG-NS + TF-IDF	100	0.6968	0.6965	0.7624	31.85	19.89	0.0000

As shown in Table 2, the manual *TF-IDF* representation achieves the best overall predictive performance among the alternatives evaluated. This result is consistent with the nature of *Sentiment140*, which consists of short texts in which the presence of certain terms can be highly discriminative for sentiment classification. In addition, constructing the TF-IDF representation is computationally very efficient, although it produces a high-dimensional and extremely sparse matrix.

The manual LSA representation obtains slightly lower predictive results than TF-IDF. Nevertheless, it reduces dimensionality considerably and generates a much denser representation. This reduction substantially decreases the time required to train the classifier. Therefore, LSA provides an interesting balance between predictive performance and representation compactness, although its initial construction is more computationally expensive because of the matrix decomposition.

By contrast, manual LDA performs substantially worse than the other representations evaluated. This result can be explained by the characteristics of the corpus, in which the documents are very short and noisy tweets that do not always contain sufficient context for a probabilistic topic model to identify stable thematic structures. Although LDA generates a very compact representation and allows the classifier to be trained quickly, fitting the representation is computationally expensive and its predictive capacity in this setting is limited.

A comparison of the computational performance of the different vector representations as their dimensionality increases is presented below. In particular, the average total pipeline time is analysed for the manual implementations developed in this study and compared with the equivalent implementations provided by high-level libraries, specifically *scikit-learn*.

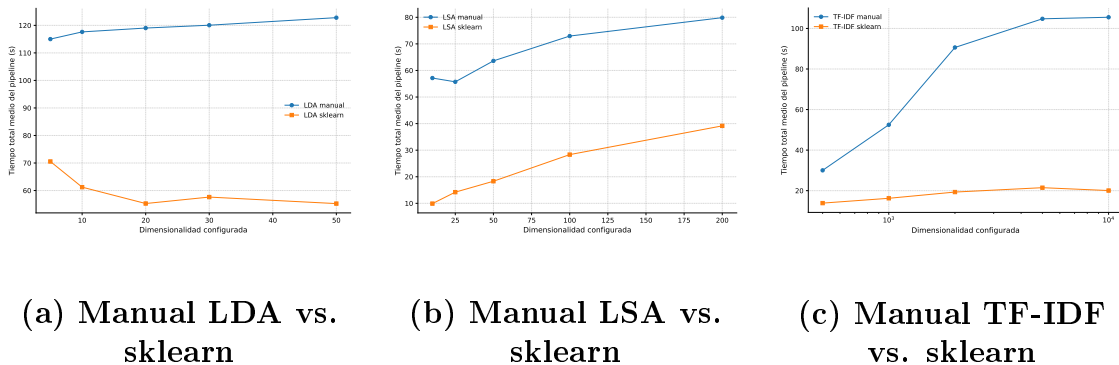


Figure 6: Comparison of the average total pipeline time between the manual implementations and the implementations provided by *sklearn*.

As shown in Figure 6, the *scikit-learn* implementations have shorter execution times than the manual versions for all three techniques analysed. This result is consistent with the use of optimised implementations and efficient matrix structures in this library (Pedregosa et al. 2011).

For *LSA*, this improvement is explained by the use of `TruncatedSVD`, which avoids performing a full matrix decomposition and can use randomised SVD or iterative methods such as ARPACK to calculate only the required latent components (Halko, Martinsson, and Tropp 2011; Lehoucq, Sorensen, and C. Yang 1998).

For *LDA*, the *scikit-learn* implementation uses variational inference in either *batch* or *online* mode, which allows topic distributions to be approximated more efficiently than through explicit manual updates over documents, terms, and topics (David M. Blei, Andrew Y. Ng, and Michael I. Jordan 2003; Hoffman, David M. Blei, and Bach 2010).

For *TF-IDF*, the difference is mainly due to the use of sparse matrices, vectorised operations, and efficient normalisation over CSR structures, compared with explicit Python loops for calculating frequencies, inverse weights, and normalisation (Salton and Buckley 1988; Virtanen et al. 2020).

These results show that manual versions are useful for understanding the internal operation of the methods, but are less computationally efficient than specialised implementations.

Regarding the SG-NS representations weighted by TF-IDF, the results must be interpreted by considering the corpus used to learn the embedding matrix. First, the manual SG-NS representation trained on Wikipedia performs worse than the TF-IDF and LSA representations. This result may be explained by the domain difference between the embedding-training corpus and the evaluation dataset.

Although Wikipedia makes it possible to learn general distributional relationships between words, its language is more formal, structured, and encyclopaedic. Consequently, it does not necessarily capture the linguistic characteristics specific to *Twitter*, such as short messages, abbreviations, colloquial expressions, spelling mistakes, emoticons, and non-standard forms.

This limitation is reduced when the SG-NS embeddings are trained directly on the *Sentiment140* corpus itself. In this case, the corpus-specific manual SG-NS representation weighted by TF-IDF substantially improves predictive performance. This improvement indicates that the model learns distributional relationships that are better adapted to the specific domain of the tweets, more effectively capturing the lexical and contextual patterns associated with sentiment within the dataset itself.

A relevant difference is also observed from a computational perspective. The SG-NS representation trained on Wikipedia requires approximately 332 seconds to construct, whereas the version trained on *Sentiment140* requires 31.85 seconds. This reduction is related to the size of the corpus used for training and the length of the documents being processed.

It should also be noted that the computational time associated with the SG-NS representation is not directly comparable with that of TF-IDF, LSA, or LDA, particularly in the case of SG-NS trained on a Wikipedia corpus. In Table 2, the SG-NS representation time corresponds to training the embedding matrix. However, in many real-world applications, this matrix is used in a pretrained form after having

been learned from large external corpora, meaning that this cost is not always part of the experimental classification pipeline.

In such cases, the effective representation time would mainly correspond to calculating the document-level aggregation, for example through a TF-IDF-weighted mean. Therefore, although SG-NS has a high learning cost when trained from scratch, its use with pretrained embeddings may be computationally more efficient in subsequent stages.

Interpretability of the TF-IDF representation Figure 7 shows the two-dimensional projection of the documents represented using TF-IDF after applying PCA. From an interpretability perspective, this visualisation makes it possible to examine whether there are principal directions of variability capable of organising the documents differently according to their class.

However, the first two principal components explain a very small proportion of the total variance, with both components jointly accounting for less than 1% of the variability in the TF-IDF matrix.

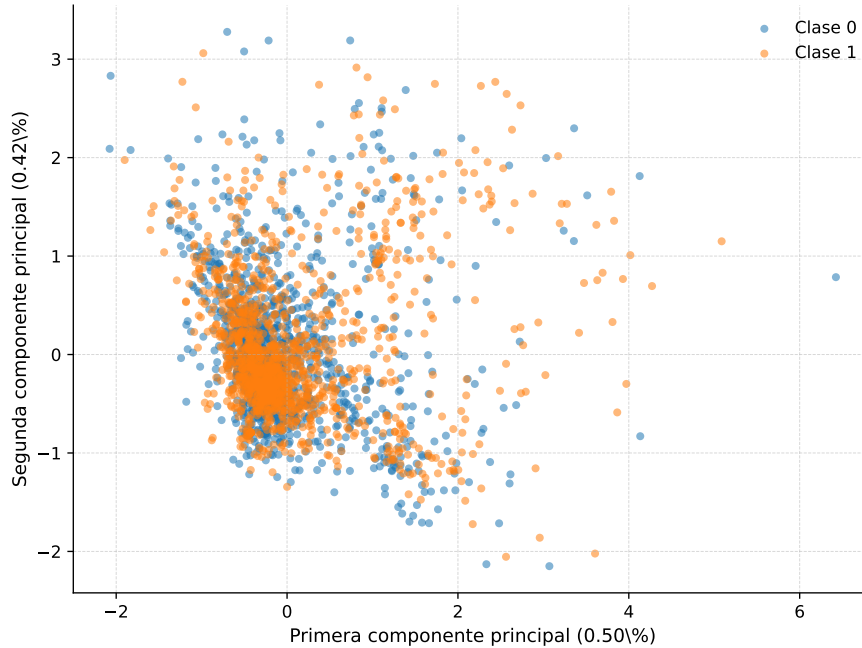


Figure 7: Two-dimensional projection of the TF-IDF space using PCA for the *Sentiment140* dataset. Each point represents a document from the training set, and its position is determined by the first two principal components calculated from the TF-IDF matrix.

This result indicates that PCA does not identify a dominant or privileged direction that concentrates a substantial part of the information in the TF-IDF space. This is consistent with the nature of this representation because TF-IDF generates very high-dimensional and highly sparse matrices in which textual information is distributed across many terms. Therefore, the variability of the corpus cannot be clearly summarised by a small number of principal components, but is instead distributed across multiple dimensions.

In addition, the two-dimensional projection shows substantial overlap between documents from class 0 and class 1. This suggests that the first two principal components

do not adequately capture the discriminative structure associated with the target variable. In other words, although TF-IDF may contain useful information for classification, this information is not concentrated in the first directions of maximum variance obtained through PCA.

A set of terms adapted to the corpus and considered especially relevant to the analysis of TF-IDF interpretability is subsequently projected into a two-dimensional space. These terms are: *good*, *great*, *happy*, *love*, *perfect*, *bad*, *sad*, *hate*, *awful*, *sucks*, *not*, *never*, *lol*, and *miss*.

The selection of these words makes it possible to analyse the behaviour of terms with positive and negative polarity, informal expressions, and elements capable of altering the semantic orientation of the message, such as negations. In addition, the cosine-similarity matrix between these terms is calculated to assess the extent to which the vector representation brings together words with similar semantic or distributional patterns within the corpus.

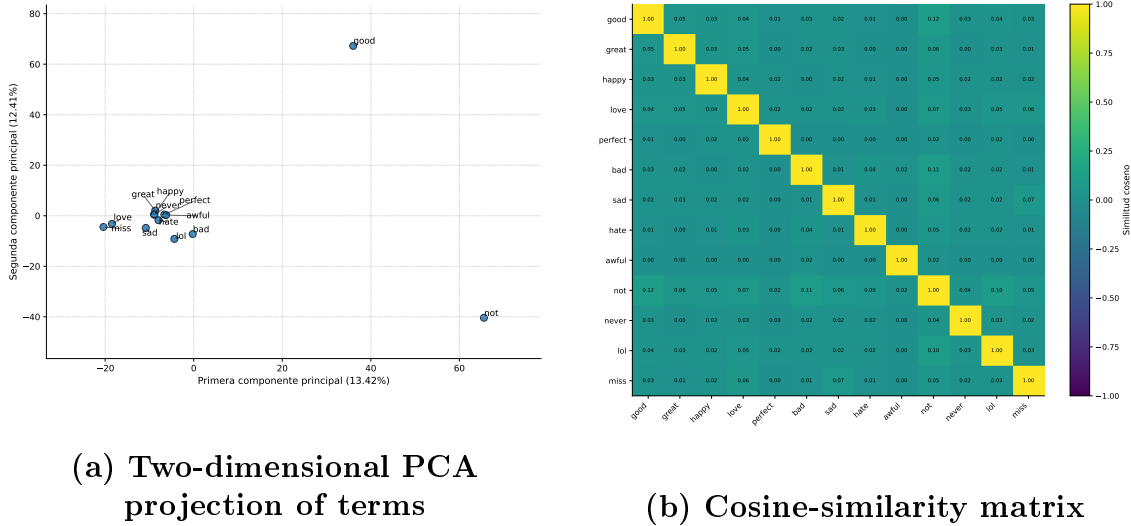


Figure 8: Analysis of representative *Sentiment140* terms in the TF-IDF space. Subfigure (a) shows the two-dimensional PCA projection of the vectors associated with the terms, whereas subfigure (b) shows the cosine similarity between those terms.

The two-dimensional projection and the cosine-similarity matrix show behaviour that is highly characteristic of this representation. First, the PCA projection shows that most of the selected terms are concentrated in a region relatively close to the origin, indicating that their TF-IDF vectors do not exhibit substantial differences along the directions of greatest variance captured by PCA.

This group contains a mixture of positive terms, such as *great*, *happy*, *perfect*, and *love*, and negative terms, such as *bad*, *hate*, *awful*, and *sad*. From a mathematical perspective, this overlap means that the projections of these terms onto the first two principal components take similar values.

Because PCA constructs new directions that maximise the global variance of the data, the proximity of words with different polarities indicates that sentiment-related differences do not constitute a dominant source of variability in the TF-IDF vectors.

The cosine-similarity matrix confirms this interpretation. Most off-diagonal similarities are very low and close to zero. This indicates that, from the perspective of

TF-IDF, the selected terms do not exhibit highly similar occurrence patterns across documents.

Even words that could be considered semantically related, such as *good*, *great*, *happy*, or *love*, do not exhibit especially high similarities. This occurs because TF-IDF does not directly capture semantic relationships or synonymy, but rather patterns of term presence and weighting across documents. Therefore, two words may have similar meanings, but if they do not occur in similar documents, their cosine similarity will be low.

Interpretability of the LSA representation Figure 9 makes it possible to visually interpret the behaviour of the LSA representation from the algebraic perspective developed in the theoretical framework.

Figure 9(a) shows that the first singular values exhibit a steep decline, followed by a more gradual decrease. This pattern indicates that the first latent components concentrate a relevant portion of the information contained in the term–document matrix, whereas the subsequent components contribute progressively less variability.

This result is consistent with the interpretation of singular value decomposition, in which each singular value σ_i measures the relative importance of a latent direction. Therefore, truncating the SVD makes it possible to retain the most informative dimensions and discard those associated with smaller weights, which may capture secondary variations or textual noise.

Figure 9(b) shows the projection of the documents onto the first two latent components of LSA, that is, the projection of the LSA document vectors onto their first two components.

Documents from both classes are shown to overlap extensively, with no clear linear separation between class 0 and class 1. This behaviour is expected for a corpus such as *Sentiment140*, which consists of short and noisy texts in which the discriminative information associated with sentiment is not necessarily concentrated in the first two latent dimensions, but instead depends on specific details.

In other words, although LSA reduces dimensionality and captures global co-occurrence relationships between terms and documents, the structure associated with sentiment polarity is distributed across a larger number of components.

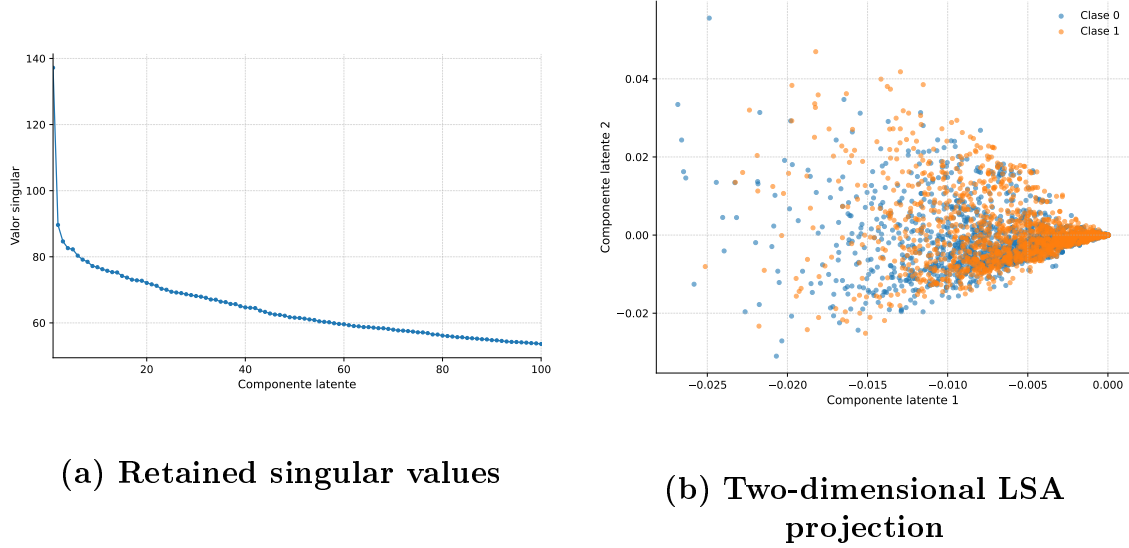


Figure 9: Analysis of the manual LSA representation for the *Sentiment140* dataset. Subfigure (a) shows the singular values retained after the matrix decomposition, whereas subfigure (b) shows the two-dimensional projection of the documents in the latent space obtained using LSA.

Projecting the previously selected terms once again into a two-dimensional space and calculating the cosine-similarity matrix between them produces the visualisation shown below.

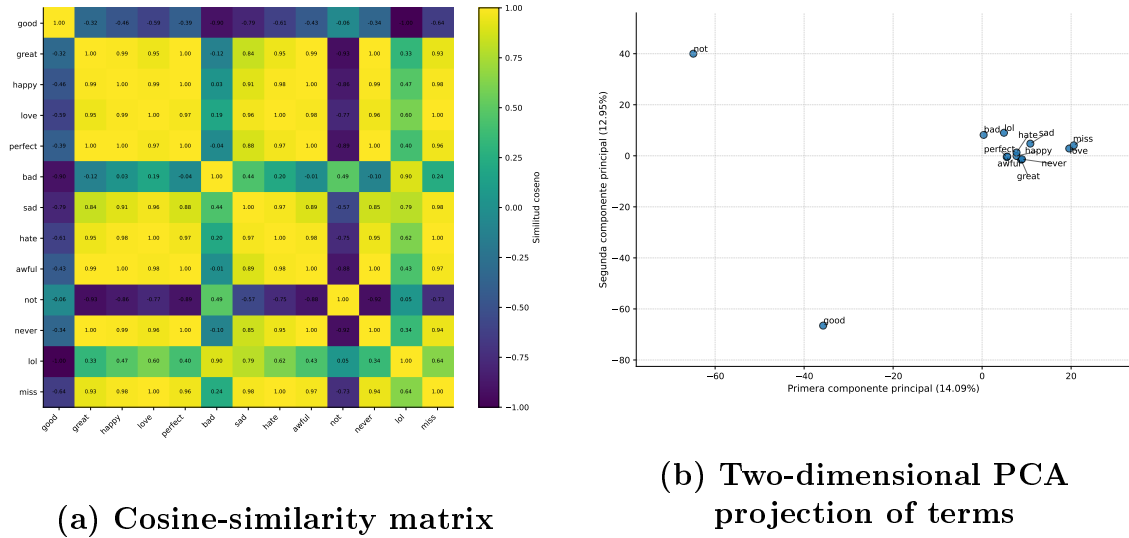


Figure 10: Analysis of representative *Sentiment140* terms in the LSA space. Subfigure (a) shows the cosine similarity between the selected terms, whereas subfigure (b) shows their two-dimensional PCA projection.

The two-dimensional projection of the terms in the LSA space exhibits behaviour very similar to that previously obtained with TF-IDF. This similarity is consistent with the nature of LSA itself, because this method begins with a term-document matrix weighted using TF-IDF and applies dimensionality reduction based on singular value decomposition.

Therefore, a representation obtained using TF-IDF followed by dimensionality reduction, such as PCA, may exhibit a geometric structure similar to that obtained

using LSA, although the two procedures are not strictly equivalent from a mathematical perspective.

The cosine-similarity matrix shows very high values between many terms that should not always be regarded as semantically equivalent. For example, positive- and negative-polarity words exhibit high similarities, such as *love* and *hate*, *happy* and *awful*, or *sad* and *miss*.

In fact, direct examination of the corpus confirmed that some word pairs with high similarity in the LSA space never occur together in any context. For example, the pairs *perfect–never* and *perfect–awful* do not co-occur in the analysed documents, but the similarity matrix assigns them relatively high values.

This result illustrates a fundamental characteristic of LSA that was established theoretically: similarity does not depend on direct co-occurrence between terms, but rather on whether those terms share similar occurrence patterns with other words and documents. Consequently, two terms may be considered close in the latent space even when they never appear together in the corpus.

Interpretability of the LDA representation In the case of LDA, it is important to bear in mind that the vector representation of each document is not interpreted as a combination of arbitrary weights, but rather as a probability distribution over latent topics.

In other words, each document is represented by a vector

$$\theta_d = (\theta_{d,1}, \theta_{d,2}, \dots, \theta_{d,K}),$$

where each component $\theta_{d,k}$ indicates the weight or proportion of topic k within document d . Therefore, the elements of the vector are non-negative and sum to one.

This property distinguishes LDA from other vector representations such as TF-IDF or LSA because each dimension in LDA has a probabilistic interpretation associated with a latent topic. Consequently, the position of a document in the LDA space should be understood as its topic mixture rather than merely as an abstract geometric coordinate.

Figure 11 shows the two-dimensional PCA projection of the documents represented by the topic distributions θ_d obtained using LDA.

From a classification perspective, substantial overlap is observed between documents from class 0 and class 1. This indicates that the topic mixtures learned by LDA do not produce a clear visual separation between the two classes in the first two principal components.

Therefore, although LDA captures some thematic organisation within the corpus, this structure does not appear to correspond directly to the target variable, or at least is not clearly reflected in this two-dimensional projection.

The vertical accumulation of points around $x = 0$ should be interpreted with caution. This line does not represent a decision boundary or a particular topic, but instead corresponds to documents whose value on the first principal component is very close to zero.

Because PCA centres the data before calculating the principal components, a value close to zero on the first component indicates that these documents lie close to the corpus average with respect to the principal axis of thematic variation.

In practical terms, this suggests that many documents exhibit similar topic mixtures along the direction captured by the first component. In other words, they are not particularly differentiated according to the main thematic contrast identified by PCA.

This observation is also consistent with the classification results, in which LDA exhibits very limited discriminative ability. If a large proportion of the documents share similar topic distributions and the classes overlap extensively in the projected space, it is reasonable to expect the topic-based representation to provide little useful information for distinguishing between the target categories.

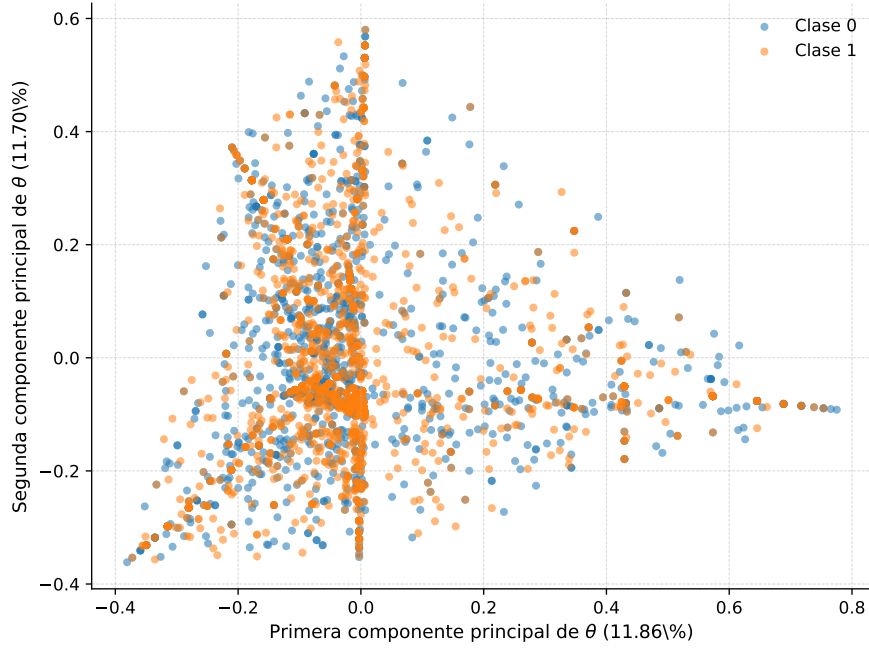


Figure 11: Two-dimensional PCA projection of the documents represented using LDA for the *Sentiment140* dataset. Each document is represented by its topic-distribution vector θ_d .

Interpretability of the SG-NS representation Figure 12 shows the two-dimensional PCA projection of the documents represented using TF-IDF-weighted SGNS embeddings. In this case, each document is not represented by a sparse term vector, but by a dense aggregation of its word embeddings, where TF-IDF weighting assigns greater relevance to the most informative terms in the document. This strategy is consistent with the theoretical formulation of document representation based on embeddings, in which the word vectors learned through Skip-gram are combined using a weighted aggregation function to obtain a fixed-dimensional document vector.

From an interpretability perspective, the PCA projection shows that the embedding-based representation organises the documents in a relatively compact and structured space. However, it should be taken into account that the SGNS embeddings used were previously trained on a corpus of Wikipedia documents and subsequently applied to the *Sentiment140* dataset, which consists of texts obtained from *Twitter*. This domain difference may affect the representational capacity of the model, because Wikipedia language tends to be more formal, structured, and stable, whereas *Twitter* language is characterised by greater informality, brevity, abbreviations, colloquial expressions, and lexical variability. Therefore, although the embeddings may

capture general semantic relationships between words, they may not represent the nuances of social-media language with the same degree of accuracy.

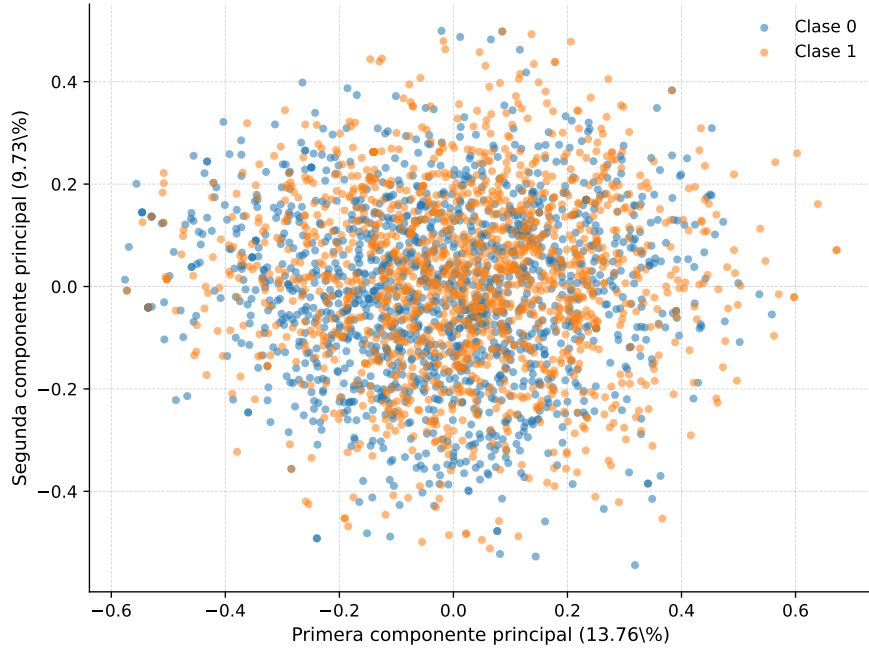


Figure 12: Two-dimensional PCA projection of the documents represented using TF-IDF-weighted SGNS embeddings. Each point represents a document from the training set projected onto the first two principal components.

Figure 13 subsequently presents a two-dimensional PCA projection of the selected terms together with the cosine-similarity matrix calculated between them. As stated previously, it should be taken into account that this projection was obtained from embeddings trained on a corpus of Wikipedia articles.

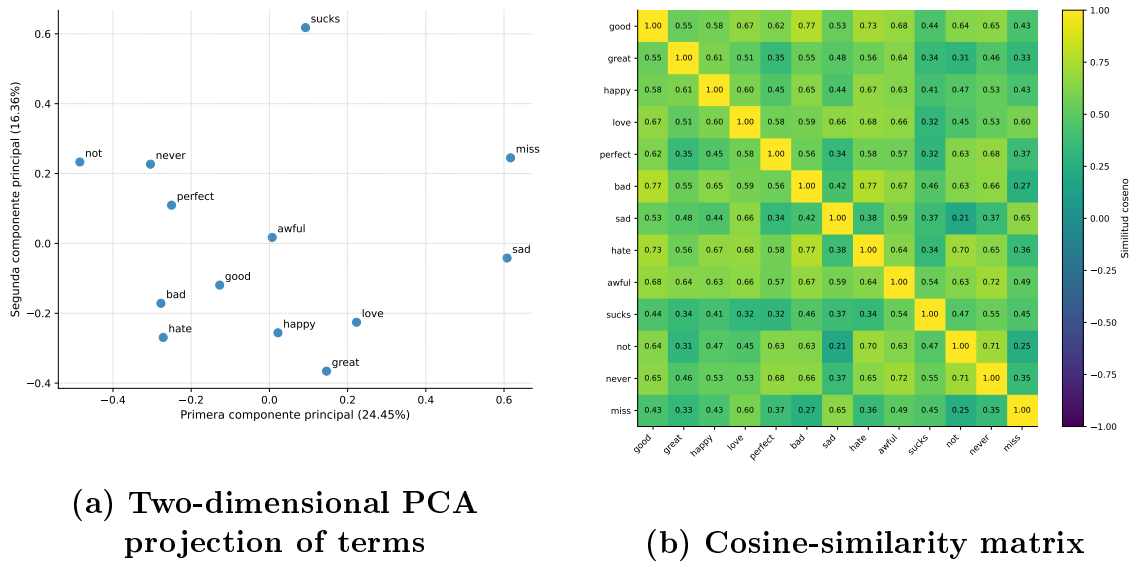


Figure 13: Analysis of representative *Sentiment140* terms in the SGNS embedding space. Subfigure (a) shows the two-dimensional PCA projection of the vectors associated with the terms, whereas subfigure (b) shows the cosine similarity between them.

The figure shows that the selected terms are not strictly grouped according to their sentiment polarity. Some positively charged words, such as *good*, *great*, *happy*, *love*, and *perfect*, do not form a single clearly differentiated group. Similarly, negative terms such as *bad*, *hate*, *sad*, *awful*, and *sucks* are not concentrated in the same region of the plane. This result is consistent with the nature of embeddings, because SG-NS learns distributional similarities from the contexts in which words occur rather than from supervised sentiment categories.

A relevant aspect is that some functionally related terms do appear close to one another. For example, *not* and *never* are located relatively close together, which can be interpreted as a proximity associated with their linguistic role as negation markers. Some proximity can also be observed among evaluative or emotional terms such as *bad*, *hate*, *happy*, *love*, and *good*. However, this proximity does not imply that the words have the same polarity, but rather that they may share usage contexts related to evaluative expressions, opinions, or emotional states.

The position of some more distant terms is also informative. For example, *sucks* appears separated from the other words, which may be due to its being a more informal or colloquial expression whose distribution in Wikipedia is less representative than it would be in a *Twitter* corpus. Similarly, the positions of *miss* and *sad* should be interpreted with caution because the embeddings were trained on Wikipedia and may therefore reflect more general or encyclopaedic uses of these words rather than the affective nuances they may acquire in short social-media messages.

For the embeddings learned from the dataset itself, Figure 14 shows a structure that is better adapted to the domain under analysis, since the words exhibit high cosine similarities and capture relationships specific to the language used in the corpus.

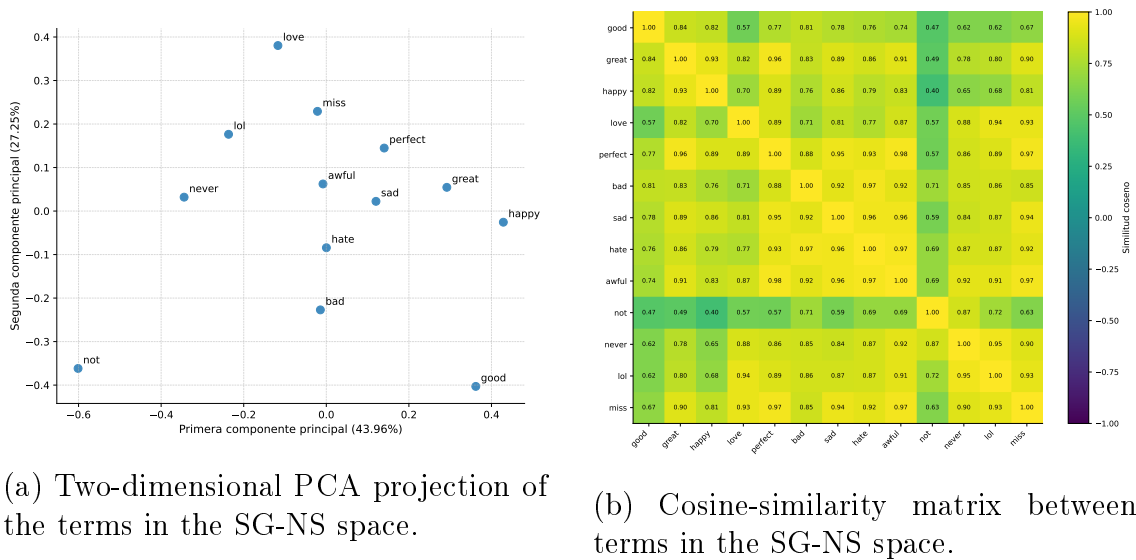


Figure 14: Visualisation of the term structure in the SG-NS embedding space using a cosine-similarity matrix and a two-dimensional PCA projection.

The two-dimensional projection of the SG-NS embeddings trained on the *Sentiment140* corpus itself shows an organisation that depends more directly on the contextual use of words within the dataset.

Unlike embeddings trained on Wikipedia, this representation is better adapted to the short and informal language of *Twitter*. For example, *lol* appears integrated

into the upper-left region, relatively close to terms such as *never*, *miss*, and *love*, suggesting that the model has captured usage patterns specific to the corpus rather than only general semantic relationships. Similarly, *miss* and *love* are located in the upper part of the plane, reflecting a possible association with emotional contexts that occur frequently in the analysed messages.

Some terms with a specific function or usage are also clearly displaced. In particular, *not* appears isolated in the lower-left region, indicating that its occurrence pattern in the corpus differs from that of the other terms. This is consistent with the fact that negations may appear in many different contexts and alter the meaning of positive or negative expressions. By contrast, *good* appears far away in the lower-right region, showing that its contextual use within the corpus does not necessarily coincide with that of other positive terms such as *happy*, *great*, *love*, or *perfect*.

The negative terms *bad*, *hate*, and *awful* appear relatively close to one another in the central region, suggesting that the model identifies a certain distributional relationship between them. However, they do not form a completely separate block from the other terms, since words such as *sad* and *perfect* also appear close to this central region. This confirms that SG-NS embeddings do not organise words exclusively by sentiment polarity, but rather according to the contexts in which they occur within the corpus.

4.5.2 Results for the arXiv Dataset

This subsection presents the results obtained for the *arXiv* dataset. This corpus makes it possible to evaluate the behaviour of the different vector representations in a binary classification setting based on scientific documents, which are characterised by greater length, a more stable technical vocabulary, and a more clearly defined thematic structure than that observed in short and informal texts. The results achieved by each method are analysed below in terms of classifier predictive performance, the computational cost associated with constructing the representation, the resulting dimensionality, and the interpretability of the generated vectors.

Predictive performance of the representations The results obtained for the *arXiv* dataset show more stable and competitive behaviour among the three manual representations than was observed for *Sentiment140*. In this case, TF-IDF, LSA, and LDA achieve high and relatively similar predictive performance, suggesting that the corpus exhibits a more regular and richer textual structure that facilitates the extraction of discriminative information. This is consistent with the nature of *arXiv*, where the documents are longer, contain a more consistent technical vocabulary, and provide greater semantic context.

Table 3: Predictive and computational results obtained for the *arXiv* dataset using manual implementations of the vector representations.

Representation	Dim.	Accuracy	F1	AUC-ROC	Repr. time (s)	Clf. time (s)	Sparsity
Manual TF-IDF	5000	0.9200	0.9199	0.9692	1.97	49.22	0.9878
Manual LSA	100	0.9193	0.9192	0.9681	157.27	15.63	0.0000
Manual LDA	10	0.9118	0.9117	0.9647	1923.32	3.06	0.0000
SG-NS Wiki + TF-IDF	100	0.8785	0.8785	0.9444	332.28	28.10	0.0000
SG-NS arXiv + TF-IDF	100	0.9255	0.9255	0.9730	220.15	20.41	0.0000

Compared with the *Twitter* dataset, the differences between the representations are

less pronounced. Whereas frequency-based techniques, particularly TF-IDF, were more suitable for capturing short terms directly associated with sentiment in *Sentiment140*, latent representations also perform effectively on *arXiv*. In particular, LSA maintains performance very close to TF-IDF, indicating that dimensionality reduction preserves a large proportion of the relevant information. LDA also becomes clearly more useful than in the *Twitter* case, because scientific documents provide sufficient length and thematic coherence for the model to identify more informative topic mixtures.

From a computational perspective, a tendency similar to that observed previously is maintained: TF-IDF is the most efficient representation to construct, whereas LSA and especially LDA require a higher fitting cost. However, once the latent representations have been obtained, classifier training and prediction are faster because of the lower dimensionality of the vectors. In this regard, it is especially useful to compare the total average time required to fit or construct each representation with the computational time subsequently required by the classifier. This comparison distinguishes between the cost associated with the text-representation stage and the cost associated with the supervised stage, showing that a representation that is more expensive to construct may subsequently reduce the computational burden of the predictive model.

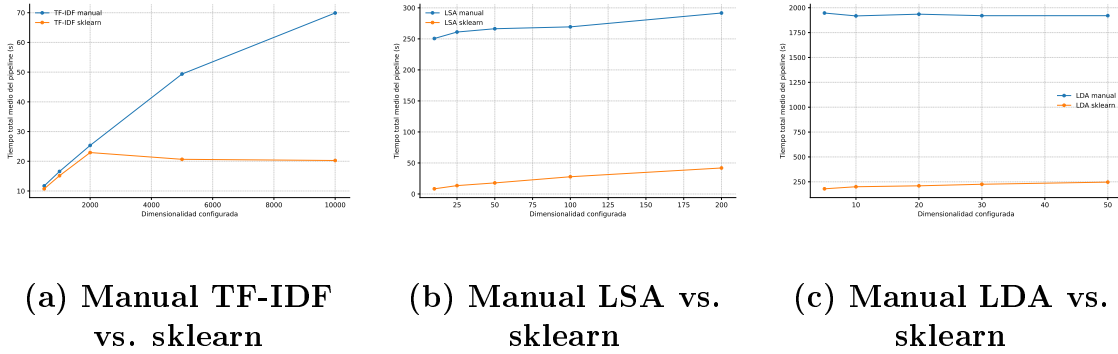


Figure 15: Comparison of the average total pipeline time between the manual implementations and the implementations provided by *sklearn* for the *arXiv* dataset.

As was also observed for *Sentiment140*, the implementations based on *scikit-learn* exhibit clearly higher computational efficiency than the manual implementations.

Regarding the representations based on *SG-NS*, a clear difference is observed depending on the corpus used to train the embeddings. The *SG-NS Wiki + TF-IDF* representation performs worse than the other main techniques, which may be explained by the mismatch between the general domain of Wikipedia and the specific technical language of *arXiv*. By contrast, when the embeddings are trained directly on the *arXiv* corpus itself, the *SG-NS arXiv + TF-IDF* representation achieves the best overall performance in the table. This result suggests that, for scientific documents, training embeddings on a corpus from the same domain makes it possible to capture the terminological relationships relevant to classification more effectively.

Interpretability of the TF-IDF representation The PCA projection of the TF-IDF representation shown in Figure 16 reveals a relatively clear separation between the two classes, especially along the first principal component. Class 0 is mostly concentrated in the left-hand region of the plane, whereas class 1 is shifted

towards positive values of the first component. Although there is a central region of overlap, the overall distribution indicates that TF-IDF captures lexical patterns that are relevant for distinguishing between *cs* and *non-cs* documents.

The second principal component, which explains a smaller proportion of the variance, appears to reflect internal variability within each class rather than a direct separation between them. Therefore, despite the high dimensionality and sparsity of TF-IDF, the two-dimensional projection retains a discriminative structure that is consistent with the strong predictive performance obtained.



Figure 16: Two-dimensional PCA projection of the documents in the *arXiv* dataset represented using TF-IDF.

Figure 17 subsequently presents the two-dimensional projection of a selection of representative terms from the *arXiv* dataset: *algorithm*, *learning*, *neural*, *network*, *data*, *model*, *classification*, *optimization*, *quantum*, *galaxy*, *star*, *mass*, *energy*, and *theorem*. The purpose of this analysis is to evaluate the ability of the representation to identify relationships between words associated with different thematic domains. In particular, the aim is to observe whether terms linked to related fields, such as machine learning, computing, physics, or astronomy, tend to occupy nearby regions of the projected space. The cosine-similarity matrix between these terms is also presented to complement the visual analysis and quantify the degree of proximity captured by the representation.

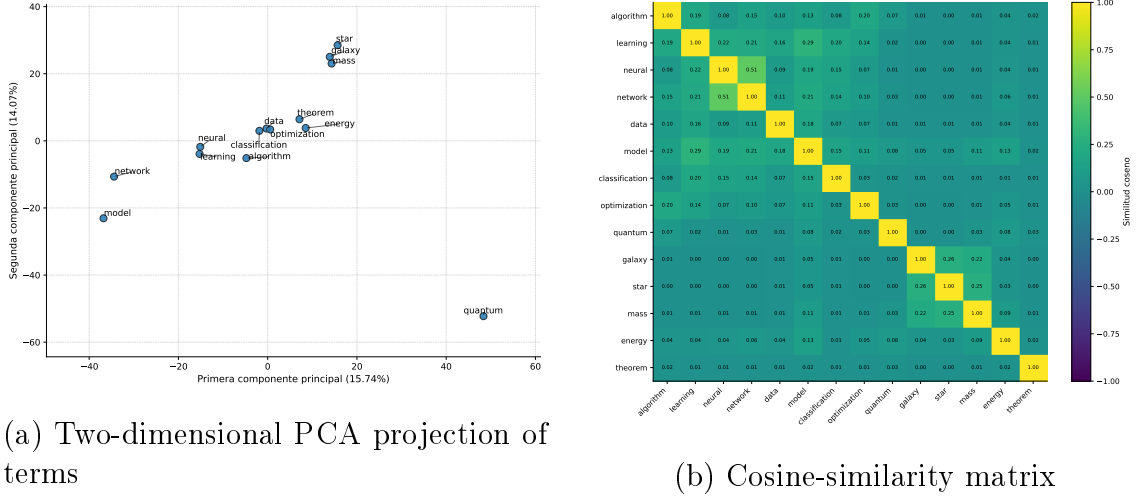


Figure 17: Analysis of representative terms from the *arXiv* dataset in the TF-IDF space. Subfigure (a) shows their two-dimensional PCA projection, whereas subfigure (b) shows the cosine-similarity matrix between the selected terms.

In the TF-IDF representation, the cosine-similarity matrix shows that the associations between terms are generally moderate or low, which is consistent with the sparse nature of this representation. TF-IDF does not directly capture semantic relationships, but rather term-occurrence patterns across documents. Consequently, the strongest similarities appear between words that are likely to share frequent document contexts, such as *neural* and *network*, or between astrophysical terms such as *galaxy*, *star*, and *mass*. By contrast, terms belonging to very different domains, such as *quantum*, *theorem*, or machine-learning terms, frequently exhibit similarities close to zero.

The two-dimensional PCA projection shows an organisation very similar to that subsequently obtained using LSA. Astronomy-related terms such as *galaxy*, *star*, and *mass* are grouped in the upper part of the figure, whereas computational terms such as *learning*, *neural*, *classification*, *algorithm*, *data*, and *optimization* occupy a more central region. The term *quantum* also appears clearly isolated, indicating that its occurrence pattern in the documents differs from that of the other selected terms. In addition, *model* and *network* are shifted towards the left, reflecting specific usage patterns within the corpus.

The fact that the TF-IDF projection is very similar to the one obtained using LSA is methodologically consistent. LSA is constructed precisely from a TF-IDF-weighted term–document matrix and applies algebraic dimensionality reduction through SVD. In this case, applying PCA directly to the TF-IDF term vectors also seeks a lower-dimensional representation that preserves the main directions of variability in the original matrix. Therefore, although PCA on TF-IDF and LSA are not exactly the same procedure, both exploit the same term–document co-occurrence structure, which explains why they produce a similar spatial organisation.

Nevertheless, it should be noted that the TF-IDF similarity matrix generally contains lower values than those observed for LSA. This is because TF-IDF retains a more direct and sparse representation of occurrence patterns, whereas LSA tends to smooth the original matrix and group related terms into shared latent dimensions.

Interpretability of the LSA representation The comparison of the singular values obtained using LSA reveals relevant differences between *Sentiment140* and *arXiv*. In both cases, an initially steep decline is observed, indicating that the first latent components concentrate a substantial proportion of the information contained in the term–document matrix. However, this decline is more pronounced for *arXiv*, where the first singular values are clearly larger and decrease sharply across the first components.

This behaviour suggests that the *arXiv* dataset presents a more clearly defined latent structure. Because the corpus consists of longer scientific documents with technical vocabulary and greater thematic coherence, the first LSA components can capture dominant semantic patterns more clearly. Therefore, the decomposition identifies latent directions with greater weight, possibly associated with thematic differences between *cs* and *non-cs* documents.

The two-dimensional projection of the documents in the LSA space also shows a more differentiated structure between classes than was previously observed for *Sentiment140*. In the case of *arXiv*, although there is still some overlap between the classes, a clearer separation is visible in the latent space, especially along the first components. This suggests that LSA captures semantic patterns that are relevant for distinguishing between documents belonging to the *cs* class and *non-cs* documents.

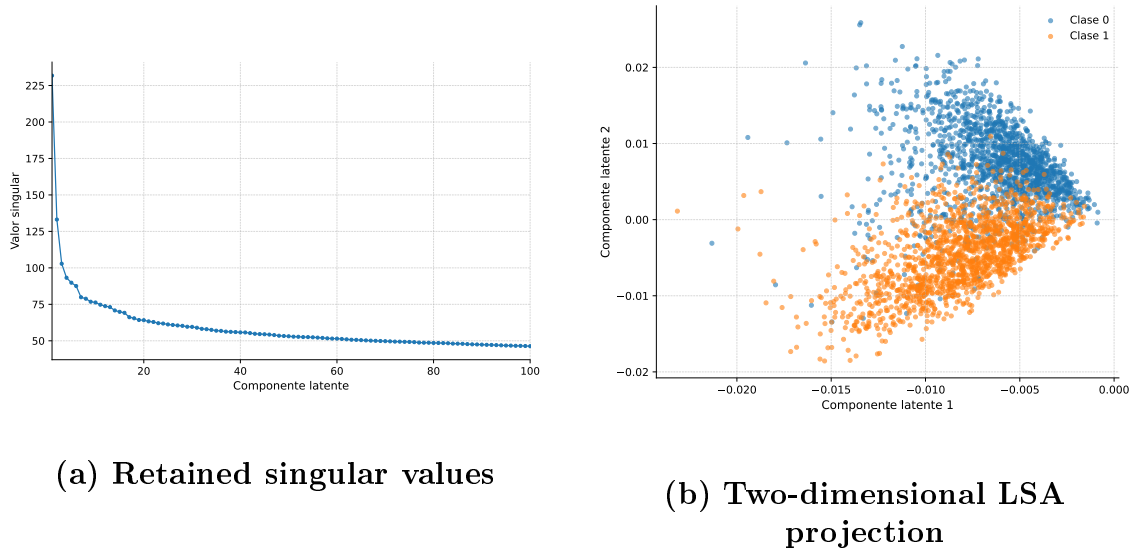


Figure 18: Analysis of the manual LSA representation for the *arXiv* dataset. Subfigure (a) shows the singular values retained by the matrix decomposition, whereas subfigure (b) shows the two-dimensional projection of the documents in the latent space generated using LSA.

Figure 19 subsequently presents the two-dimensional projection of the terms selected previously and the cosine-similarity matrix calculated between them.

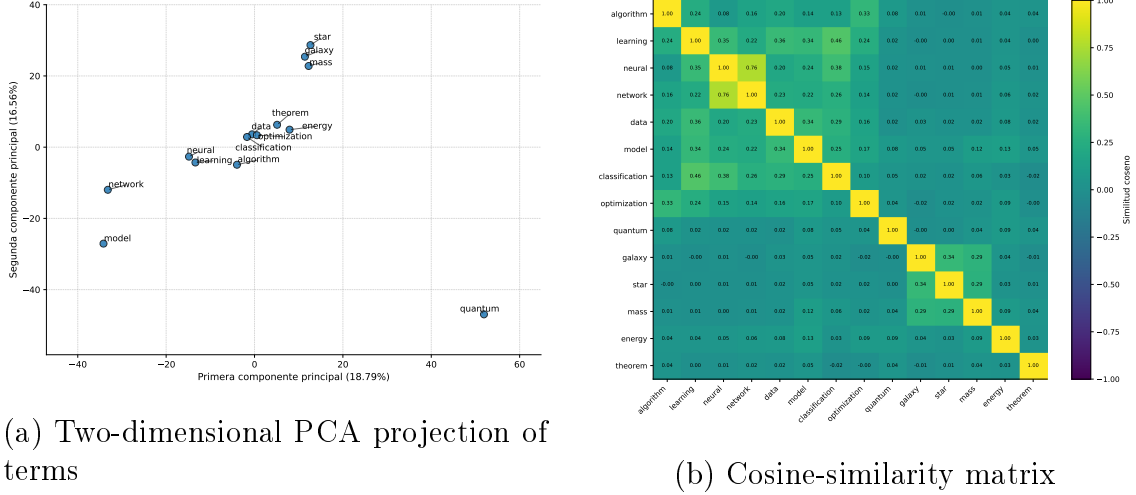


Figure 19: Analysis of representative terms from the *arXiv* dataset in the LSA space. Subfigure (a) shows the two-dimensional PCA projection of the selected terms, whereas subfigure (b) shows the cosine similarity between them in the latent space.

The cosine-similarity matrix shows that LSA captures several coherent thematic relationships between the selected terms. For example, *neural* and *network* exhibit high similarity, which is expected because they frequently occur together in machine-learning documents. Moderate associations can also be observed among *learning*, *classification*, *data*, *model*, and *optimization*, indicating that LSA partially groups vocabulary related to the computational domain.

A second thematic group is formed by terms such as *galaxy*, *star*, and *mass*, which exhibit positive similarities and appear close to one another in the two-dimensional projection. This grouping suggests that the LSA representation is capable of detecting latent relationships associated with different scientific domains, in this case astronomy or astrophysics.

In the PCA projection, the first two components jointly explain a relevant proportion of the variability in the latent space. The terms are not randomly distributed, but instead appear organised into differentiated regions: on the one hand, terms related to computing and machine learning; on the other, terms associated with physical or astronomical phenomena. This reinforces the idea that LSA extracts a certain thematic structure from term–document co-occurrence patterns.

Nevertheless, some terms exhibit more isolated behaviour. In particular, *quantum* is located far away from the other terms in the projection, indicating that its latent occurrence pattern in the corpus differs from that of the other selected terms. Similarly, *model* and *network* are shifted towards the left-hand region, suggesting that, although they belong to the computational domain, they exhibit specific directions of variability within the LSA space.

Interpretability of the LDA representation Figure 20 shows the two-dimensional PCA projection of the documents represented by their topic distributions θ_d obtained using LDA. In this case, a marked geometric structure is observed, in which the first principal component appears to act as an axis of thematic differentiation: class 1 documents are mainly concentrated in the left-hand region of the plane, whereas class 0 documents extend more strongly towards the right.

This distribution suggests that LDA has learned topic mixtures with a certain discriminative capacity. Class 1 appears more compact, indicating a more homogeneous thematic composition, whereas class 0 exhibits greater dispersion, reflecting greater thematic heterogeneity. Nevertheless, there is also a central region of overlap between the two classes, which may correspond to documents with similar or more ambiguous topic mixtures.

The fan-shaped point cloud is consistent with the probabilistic nature of LDA, because each document is represented as a topic distribution whose weights are non-negative and sum to one. Therefore, the documents do not freely occupy the entire vector space, but instead are organised according to the constraints inherent to a thematic mixture. In addition, the second principal component appears to capture internal variability, especially within class 0, where different thematic profiles can be observed.

In this case, the projection indicates that LDA is more informative for this dataset than for *Sentiment140*, since the greater length and thematic coherence of the documents make it possible to learn topic distributions that are more useful for distinguishing between classes.

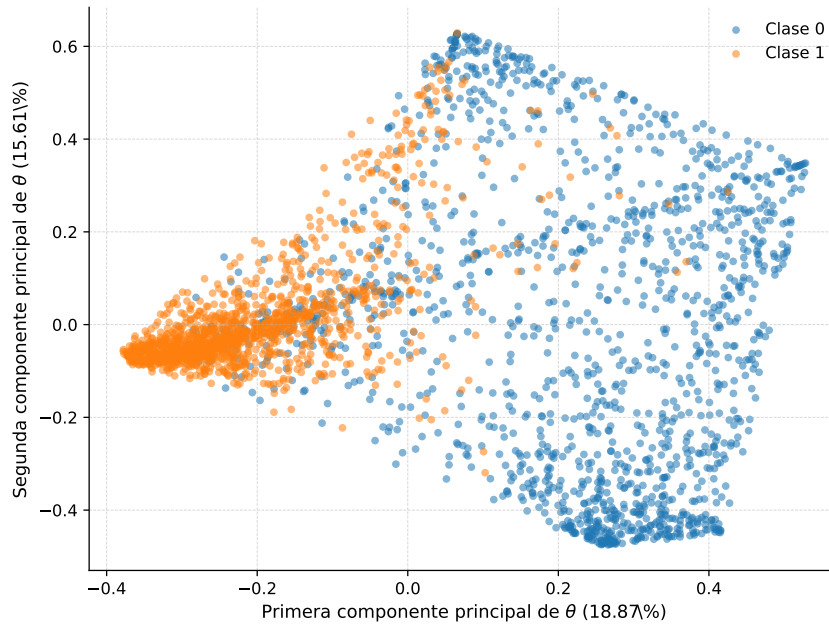
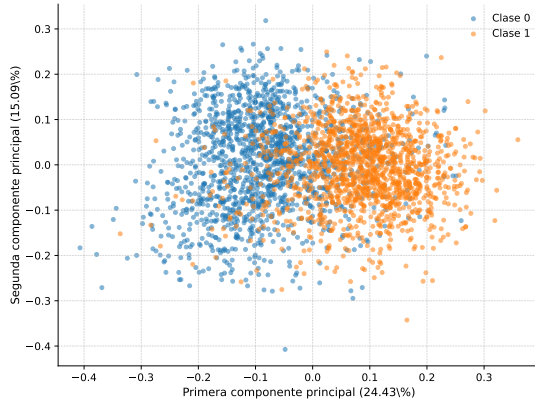


Figure 20: Two-dimensional PCA projection of the documents represented by their topic distributions in the manual LDA model for the *arXiv* dataset.

Interpretability of the SG-NS representation Figure 21 compares the PCA projections of the document representations obtained using TF-IDF-weighted SGNS embeddings. In both cases, each document is represented as a dense vector constructed from its term embeddings, weighted according to their TF-IDF relevance.



(a) SGNS trained on Wikipedia

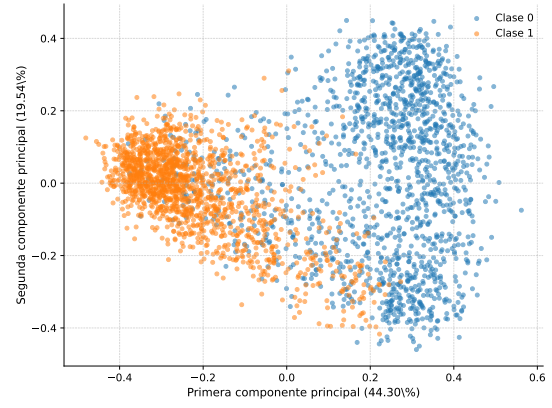
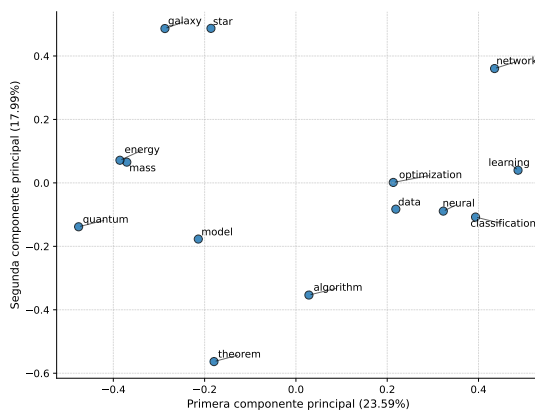
(b) SGNS trained on *arXiv*

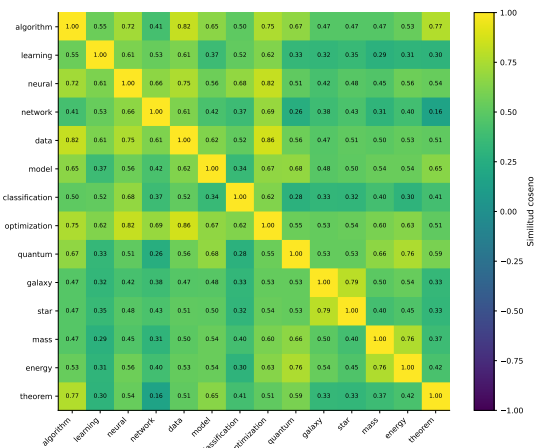
Figure 21: Two-dimensional PCA projection of the *arXiv* documents represented using SGNS embeddings aggregated through a TF-IDF-weighted mean. Subfigure (b) shows the results obtained with embeddings trained on the *arXiv* corpus itself, whereas subfigure (a) shows the results obtained with embeddings trained on Wikipedia.

When comparing the two projections, the embeddings trained on the *arXiv* corpus itself produce a clearer separation between the classes than the embeddings trained on Wikipedia. In the Wikipedia-based representation, there is considerable overlap between class 0 and class 1 documents, whereas in the representation trained on *arXiv*, the classes occupy more differentiated regions of the plane. In addition, the first principal component explains a larger proportion of the variance for the embeddings trained on *arXiv*, suggesting that they capture patterns specific to the analysed scientific domain more effectively.

Figure 22 shows, on the one hand, the two-dimensional PCA projection of the embeddings associated with the selected terms and, on the other, the cosine-similarity matrix between them.



(a) Two-dimensional PCA projection of terms



(b) Cosine-similarity matrix

Figure 22: Analysis of representative terms from the *arXiv* dataset in the space of SG-NS embeddings trained on Wikipedia. Subfigure (a) shows the two-dimensional PCA projection of the selected terms, whereas subfigure (b) shows the cosine-similarity matrix between them.

The projection shows the presence of several relatively differentiated groups of words. First, the terms *galaxy* and *star* appear very close to one another, which is consistent with their being related concepts in astronomy. Similarly, terms such as *energy*, *mass*, and *quantum* occupy nearby regions, reflecting relationships associated with the domain of physics. A coherent grouping of terms linked to machine learning and computing can also be observed, including *optimization*, *learning*, *data*, *neural*, and *classification*. These results suggest that the SG-NS embeddings trained on Wikipedia are capable of reasonably capturing general semantic relationships between terms belonging to different scientific domains.

However, one aspect that must be taken into account is that the PCA projection should be interpreted with caution. The first two principal components explain approximately 41% of the total variance, which is a relevant but not exhaustive proportion. Consequently, approximately 59% of the variability in the original space is not represented in this two-dimensional visualisation. This implies that some relationships present in the complete embedding space may be partially distorted or may not be clearly reflected in the projection.

An example of this can be observed when comparing the positions of *energy*, *mass*, and *quantum*. PCA places *energy* and *mass* close together, and the cosine-similarity matrix confirms that *energy* is one of the terms most similar to *mass*. However, the matrix also shows a relevant relationship between *mass* and *quantum*, which is not equally clear in the two-dimensional projection. Therefore, although PCA is useful for obtaining a global and interpretable view of the vector space, it must be complemented with metrics calculated in the full-dimensional representation, such as cosine similarity.

Figure 23 subsequently presents the projection of the SG-NS embeddings trained on the *arXiv* dataset itself for the terms selected previously.

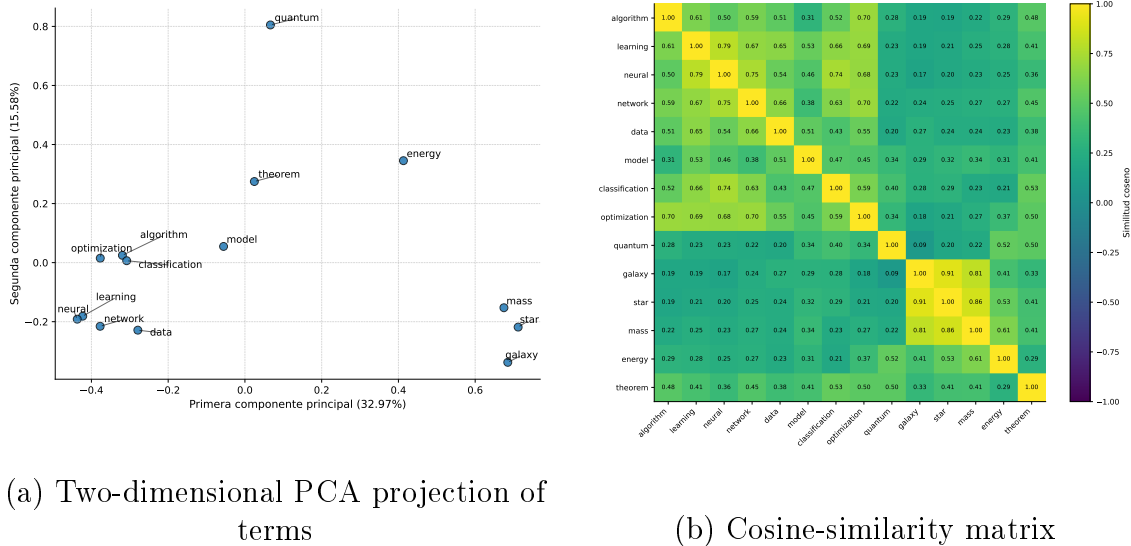


Figure 23: Analysis of representative terms from the *arXiv* dataset in the space of SG-NS embeddings trained on the training set itself. Subfigure (a) shows the two-dimensional PCA projection of the selected terms, whereas subfigure (b) shows the cosine-similarity matrix between them.

For the embeddings learned from *arXiv*, the organisation is more closely adapted to the scientific domain under analysis. Astronomy-related terms such as *galaxy*

and *star* appear very close to one another, whereas physical concepts such as *mass*, *energy*, and *quantum* also exhibit clear relationships in the cosine-similarity matrix. Similarly, terms associated with machine learning, such as *learning*, *neural*, *network*, *classification*, and *optimization*, form a coherent group.

The embeddings trained on *arXiv* are especially notable for capturing stronger and more pronounced similarities, reflecting relationships between concepts belonging to the same scientific domain more clearly. Because they were trained on a specialised corpus, the representations tend to concentrate semantic information in more specific associations, producing well-defined clusters and higher similarity values between related terms. By contrast, the embeddings trained on Wikipedia also identify coherent semantic relationships, but these appear in a smoother and more diffuse form because of the broader and more heterogeneous nature of the training corpus. As a result, the similarities are distributed across a greater number of concepts and contexts, providing more general-purpose but less specialised representations than those obtained from the *arXiv* corpus.

5 Conclusions

The main objective of this study was to analyse, formalise, and compare different text vector-representation techniques from both theoretical and empirical perspectives. In particular, statistical, algebraic, probabilistic, and neural approaches were studied, specifically *TF-IDF*, *LSA*, *LDA*, and dense representations constructed from embeddings learned through *Skip-gram with Negative Sampling (SG-NS)*. The analysis conducted leads to the conclusion that document vector representation is not merely a preliminary stage before training supervised models, but a central component that determines predictive performance, computational cost, interpretability, and the scalability of the final system.

From a theoretical perspective, the study showed that each technique is based on a different conception of textual meaning. *TF-IDF* represents documents according to the statistical importance of their terms, combining local and global information through term frequency and inverse document frequency. *LSA*, in turn, introduces an algebraic transformation of the term–document matrix using singular value decomposition, making it possible to reduce dimensionality and capture latent co-occurrence structures. *LDA* adopts a probabilistic perspective by representing each document as a distribution over latent topics. Finally, *SG-NS* learns dense word representations through a context-based predictive task, after which document vectors can be constructed using aggregation strategies.

A first relevant conclusion is that no single representation is universally superior for every corpus or task. The results show that the behaviour of each method clearly depends on the characteristics of the dataset. For *Sentiment140*, which consists of short, informal, and noisy texts, the manual *TF-IDF* representation achieves the best overall predictive performance. This result is consistent with the nature of sentiment analysis in tweets, where the presence of certain specific terms may be highly discriminative. In this setting, representations based on latent structures, especially *LDA*, encounter greater difficulties because the documents contain little context from which to infer stable topics.

Nevertheless, *LSA* achieves performance close to that of *TF-IDF* on *Sentiment140*, while substantially reducing dimensionality. Although its accuracy is slightly lower,

the resulting representation is much more compact, significantly reducing the time required to train the classifier. Therefore, *LSA* provides an interesting balance between predictive performance and the subsequent efficiency of the supervised model. This conclusion is important because it shows that a lower-dimensional representation can retain a relevant proportion of the discriminative information while simultaneously reducing the computational cost of classification.

By contrast, *LDA* exhibits clearly limited behaviour on *Sentiment140*. Although it generates highly compact and interpretable representations, its predictive performance is low compared with *TF-IDF* and *LSA*. This limitation can be attributed to the brevity and noise of the corpus, because topic models require sufficient textual context to identify coherent thematic distributions. Consequently, the probabilistic interpretability of *LDA* does not, by itself, guarantee strong predictive performance, particularly when the task is not based on clear thematic differences but on short lexical signals associated with sentiment.

The results obtained using *SG-NS* also demonstrate the importance of the embedding-training domain. When embeddings are trained on an external corpus such as Wikipedia, the resulting representation captures general semantic relationships but does not necessarily reflect the linguistic characteristics of *Twitter*. For this reason, its performance on *Sentiment140* is lower than that of *TF-IDF* and *LSA*. However, when embeddings are trained directly on the *Sentiment140* corpus itself, performance improves substantially. This indicates that dense neural representations can be more useful when the training corpus is consistent with the application domain, especially for informal texts containing expressions, abbreviations, and usage patterns that are not representatively present in more formal external corpora.

For the *arXiv* dataset, the results show different behaviour. In this case, the documents are longer, more technical, and more structured, allowing latent and thematic representations to perform more competitively. *TF-IDF*, *LSA*, and *LDA* achieve high and similar performance, with AUC-ROC values above 0.96. This confirms that, when documents contain greater semantic density and more stable terminology, techniques that exploit global co-occurrence structures or thematic distributions can capture information that is relevant to classification.

In particular, *LSA* achieves performance on *arXiv* that is practically equivalent to that of *TF-IDF*, but with much lower dimensionality. This result reinforces the usefulness of algebraic dimensionality reduction for scientific corpora, where more consistent semantic patterns and latent relationships between terms and documents exist. In addition, *LDA* improves clearly compared with its performance on *Sentiment140*, showing that its usefulness depends heavily on the existence of a sufficiently clear thematic structure. Although its fitting cost is high, its representation provides an interpretable view of documents as mixtures of topics, which may be especially valuable when explainability is a priority.

The best-performing representation evaluated on the *arXiv* dataset consists of *SG-NS* embeddings trained on the corpus itself and aggregated using *TF-IDF* weighting. This result confirms that neural embedding-based representations can benefit substantially from training on a domain-specific corpus. In scientific texts, where technical vocabulary and terminological associations are highly relevant, embeddings trained on the dataset itself capture the distributional relationships required to distinguish between *cs* and *non-cs* documents more effectively. Therefore, the correspondence between the embedding-training domain and the evaluation domain emerges as a determining factor.

From a computational perspective, the experiments show a clear distinction between the cost of constructing the representation and the cost of training the classifier. *TF-IDF* stands out for its short construction time, especially compared with *LSA* and *LDA*. However, its high dimensionality may increase classifier-training time. By contrast, representations such as *LSA* and *LDA* require a greater initial cost but generate much more compact vectors, reducing the subsequent computational burden of the supervised model. This observation shows that the efficiency of a representation should not be evaluated solely in terms of its construction time, but rather through the total pipeline cost and the relationship between dimensionality, performance, and interpretability.

It was also confirmed that the manual implementations require more time than the optimised implementations provided by libraries such as *scikit-learn*. This result was expected because specialised libraries use sparse structures, vectorised operations, and optimised algorithms. Nevertheless, the manual implementations fulfil a relevant role within this study because they establish a direct relationship between the mathematical formulation of each method and its computational operations. In this regard, the contribution of the study lies not only in obtaining the highest possible practical performance, but also in understanding and demonstrating how each representation is constructed from its foundations.

Regarding interpretability, the results make it possible to distinguish between different forms of explanation. *TF-IDF* provides a direct interpretation at the term level because each dimension is associated with a specific word in the vocabulary. However, this advantage is limited by the high dimensionality and sparsity of the resulting matrix. *LSA* reduces dimensionality and captures latent relationships, but its components are algebraic combinations of terms and are therefore less immediately interpretable. *LDA* provides a more interpretable probabilistic representation by expressing each document as a mixture of topics, although its usefulness depends on those topics being coherent and relevant to the task. Finally, *SG-NS* generates dense spaces capable of capturing distributional relationships, but its dimensions are not directly interpretable and require complementary geometric analyses such as projections, cosine similarities, or studies of the global structure of the embeddings.

With regard to the state of the art, the foundational studies on *TF-IDF*, *LSA*, *LDA*, and *SG-NS* respectively established the statistical, algebraic, probabilistic, and neural foundations of these representations (Salton, Wong, and C.-S. Yang 1975; Deerwester, Susan T. Dumais, et al. 1990b; David M. Blei, Andrew Y. Ng, and Michael I. Jordan 2003; Mikolov et al. 2013). Much of the subsequent literature analysed these techniques individually, focused on their theoretical foundations, or evaluated their performance in specific applications and corpora. The present study extended this approach by integrating all four representations into a single experimental protocol, applying common preprocessing, the same training and test partitions, a comparable classifier, and a homogeneous set of metrics across two corpora with clearly different characteristics. In addition, it did not limit itself to using implementations available in specialised libraries, but developed original implementations from the mathematical formulation of the methods and subsequently compared their results with optimised versions. Therefore, the contribution did not consist of proposing a new representation algorithm, but of advancing beyond a purely theoretical comparison or one based on closed tools by providing a reproducible evaluation that explicitly connected the equations, their computational translation, and their consequences

for performance, dimensionality, cost, and interpretability.

5.1 Limitations of the study

Despite the results obtained, this study has several limitations that should be taken into account. First, the empirical comparison was conducted on two specific datasets: *Sentiment140* and *arXiv*. Although these corpora make it possible to study clearly differentiated textual settings, the results should not be automatically generalised to every textual domain. Other datasets, such as news articles, legal texts, biomedical documents, or multilingual corpora, may exhibit different behaviour.

Second, the evaluated tasks were formulated as binary classification problems. This decision made it possible to standardise the comparison between representations, but simplified the original structure of some corpora, especially *arXiv*, which contains multiple scientific categories and subcategories. Therefore, a multiclass evaluation could provide a more complete view of the ability of each representation to discriminate between more specific thematic domains.

Another limitation is the use of a single supervised classifier to compare the representations. This decision makes it possible to isolate the effect of the vector representation more effectively, but it does not determine whether some techniques might benefit more from other predictive models.

It should also be considered that the manual implementations developed in this study prioritised methodological clarity and correspondence with the mathematical formulation rather than maximum computational efficiency. Therefore, the execution times obtained should not be interpreted as an optimal estimate of the real cost of each technique in production environments. In practical applications, the use of optimised libraries may substantially alter the observed computational performance.

For *SG-NS*, the document representation was constructed by aggregating word embeddings, primarily using *TF-IDF* weighting. Although this strategy produces fixed-dimensional dense vectors, it also entails the loss of information concerning word order and the syntactic structure of the sequence. Therefore, although the embeddings capture distributional relationships between terms, the final document representation does not explicitly model complex sequential dependencies.

Finally, two-dimensional PCA visualisations should be interpreted as exploratory tools rather than as complete evidence of separability or representational quality. A two-dimensional projection may hide relevant information present in later components or distort relationships existing in the original space. For this reason, the interpretative conclusions were complemented with quantitative classification metrics and similarity analyses, but they should not depend solely on the projection figures.

5.2 Future work

As a natural continuation of this study, a first line of future work would consist of extending the evaluation to other datasets and tasks. It would be especially interesting to incorporate corpora from specialised domains and texts in different languages in order to analyse whether the conclusions observed here remain valid in

more diverse settings. Similarly, evaluation on tasks other than classification, such as information retrieval, document clustering, or semantic search, would make it possible to assess other properties of the studied representations.

A second line of work would involve further investigating sequential architectures for constructing document representations. Although this study developed embedding learning through *Skip-gram with Negative Sampling*, it did not theoretically or experimentally introduce recurrent neural networks for representing complete documents. This extension would constitute a natural continuation of the analysis because RNNs process text as an ordered sequence of tokens rather than solely as a collection of words or an aggregation of embeddings. In this regard, future research could examine *RNN*, *LSTM*, or *GRU* models trained in a supervised or self-supervised manner to determine the extent to which explicitly incorporating word order improves document representation compared with bag-of-words methods or embedding averages.

Another line of future work would consist of comparing the analysed representations with more recent contextual models such as *BERT* or models based on *Transformer* architectures. Unlike *SG-NS*, these models generate context-dependent representations, so the same word may receive different vectors depending on the sentence in which it occurs. This comparison would extend the analysis from static embeddings to contextual representations, which currently dominate many natural language processing tasks.

Another possible extension would be to conduct a systematic hyperparameter search. In particular, it would be relevant to analyse the effect of vocabulary size, the number of latent components in *LSA*, the number of topics in *LDA*, embedding dimensionality, context-window size in *SG-NS*, and the number of negative samples. These factors can substantially alter predictive performance, interpretability, and computational cost.

In conclusion, this study shows that text vector representation is a multidimensional problem involving mathematical, computational, linguistic, and interpretative factors. The comparison conducted makes it possible to understand the advantages and limitations of different approaches more clearly and reinforces the idea that the choice of representation should not be based solely on a performance metric, but rather on a joint assessment of the corpus, the task, efficiency, and system explainability.

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